

A gene mutation is a permanent alteration in the DNA sequence that makes up a gene, such that the sequence differs from what is found in most people. Mutation ranges in size, they can affect anywhere from a single DNA building block (base pair) to a large segment of a chromosome that includes multiple genes.

99. When one gene controls two or more different characters simultaneously, the phenomenon is called :

- (a) Apomixis
- (b) Pleiotropy
- (c) Polyploidy
- (d) Polyteny

I.A.S. (Pre) 2002

Ans. (b)

Pleiotropy occurs when one gene influences multiple, seemingly unrelated phenotypic traits, an example being phenylketonuria, which is a human disease that affects multiple systems but is caused by one gene defect. Consequently, a mutation in a pleiotropic gene may have an effect on several traits simultaneously. Other well-known examples of pleiotropy include albinism and sickle cell anaemia.

100. In gene therapy, to correct the functioning of a faulty gene :

- (a) Bad gene is eliminated
- (b) Another good gene is inserted
- (c) Defective gene is stopped from working
- (d) The whole bad gene is replaced by a good gene

U.P.C.S. (Spl.) (Mains) 2004

Ans. (*)

Gene therapy is a technique for correcting defective genes responsible for disease development. Gene therapy is designed to introduce genetic material into cells to compensate for abnormal genes or to make beneficial protein. If a mutated gene causes a necessary protein to be faulty or missing, gene therapy may be able to introduce a normal copy of the gene to restore the function of the protein. Gene therapy replaces a faulty gene or adds a new gene in an attempt to cure disease. Gene therapies can work by several mechanisms:

- Replacing a mutated gene that causes disease with a healthy copy of the gene.
- Inactivating, or 'knocking out,' a mutated gene that is functioning improperly.
- Introducing a new or modified gene into the body to help treat a disease.

Miscellaneous

Notes

Scope of Biology :

- Human beings are the most intelligent organism due to the capability of thinking and understanding. For a human, the study of science has dual purposes. Firstly, they gain scientific knowledge. Secondly, they apply that knowledge for human welfare. The scope of biology can be discussed in the following points :

1. **Agriculture :** Biology gives knowledge about the useful and harmful plants. So that the farmer can get better pest free crops.
2. **Human health :** Biology provides the knowledge to maintain good health by following- proper food, exercises, good habits, etc.
3. **Human diseases :** Several pathogens cause different diseases. The habit, habitat, structure, life cycle etc. of such pathogens can be studied in biology. So, we can be away from different diseases to a considerable extent.
4. **Industries :** Most of the industries are based on the products of plants and animals i.e. wood, jute, rubber, gum, sugar, leather, etc.
5. **Conservation of flora and fauna :** Biology helps the scientists or researchers to conserve endangered flora and fauna.
6. **Animal husbandry :** Rearing of animals for food, agriculture, transportation etc. is an old practice done by man. Biology gives an idea to keep such animals in a systematic way.
7. **Entertainment or Aesthetic value :** Rearing of animals and preserving plants are done by the human for entertainment as well i.e. zoo, circus, aquarium, park, garden, etc.
8. **Economic value :** The products obtained by the rearing animals and planting can be sold in the market which gives financial aids. Tourist visit natural resources, wildlife, medicinal plants, etc. By exposing flora and fauna, the country can raise its economic status.
9. **Practical scopes :** The concept of biology is essential in veterinary science, dairy science, pisciculture, apiculture, biotechnology, biophysics, biochemistry, bioengineering, etc. Biology accepted the challenges given by the world regarding food, population, and diseases with solutions i.e. new technologies in agriculture, family planning and medicine.

Miscellaneous Important Facts :

- **Jonas Edward Salk** was an American medical researcher and virologist. He discovered and developed one of the first successful **polio vaccines**.
- **Albert Bruce Sabin** was a Polish American medical researcher, best known for developing the **oral polio vaccines**, which has played a key role in nearly eradicating the disease.
- **Robert Gordon Webster** is a virologist of New Zealand. He posited that pandemic strains of flu arise from genes in flu virus strains in non-humans; for example, via a reassortment of genetic segments (antigenic shift) between viruses in humans and non-humans (especially birds) rather than by mutations (antigenic drift) in annual human flu strains. He identified the flu virus H_5N_1 , isolated it and his work is also responsible for the method of human influenza vaccination that is commonly used.
- The most effective chemical agent (Chemical weapon) of the First World War was **Sulphur mustard** known as **Mustard Gas**. It is a volatile oily liquid. Mustard gas has extremely powerful vesicant (blistering) effects on its victims. In addition, it is strongly mutagenic and carcinogenic as well as lipophilic. It produces chemical burns on contacted skin within 24 hours of its exposure. The **Germans** used mustard gas for the first time during the war in 1917. They outfitted artillery shells and grenades with mustard gas that they fired in the vicinity of the troop target.

Some Medical Devices	
Devices	Uses
Stethoscope	To measure the heartbeat
Sphygmomanometer	An instrument for measuring blood pressure
Computed Tomography or CT scan	It is used to detect any abnormality or disorders in the whole body
Electrocardiography (ECG)	It is used to detect abnormalities related to the heart
Electroencephalogram (EEG)	It is used to record electrical activity of the brain
Autoanalyzer	An instrument to test the glucose, urea, cholesterol, etc.
Magnetic Resonance Imaging (MRI)	It is a medical imaging technique used in radiology to form pictures of the internal organs of the body

Important Days

International Yoga Day	21 June
World Malaria Day	25 April
World Cancer Day	4 February
World Tuberculosis Day	24 March
World Environment Day	5 June
World Food Day	16 October
World Animal Day	4 October
International Ozone Day	16 September
World Leprosy Day	Last Sunday of January
World Diabetes Day	14 November
World AIDS Day	1 December
World Alzheimer's day	21 September
World Health Day	7 April
World Thalassaemia Day	8 May
World Polio Day	24 October
World Arthritis Day	12 October
World Mental Health Day	10 October
World Homeopathy Day	10 April
Earth Day	22 April
World Water Day	22 March

- **Tomography** or **CT Scan** is imaging by sections or sectioning, through the use of any kind of penetrating wave like X-ray. The method is used in radiology, archaeology, biology, atmospheric science, geophysics, oceanography, plasma physics, material science, astrophysics, and other areas of science.
- **Electroencephalography (EEG)** : It is an electrophysiological monitoring method to record the electrical activity of the brain. It is typically non-invasive with the electrodes placed along the scalp.
- **Electrooculography (EOG)** : It is a technique for measuring the corneo-retinal standing potential that exists between the front and back of the human eye. The resulting signal is called the **electrooculogram**. Its primary applications are in ophthalmological diagnosis and in recording eye movements.

Nobel Prize in Physiology or Medicine

- The Nobel Prize in Physiology or Medicine is awarded yearly by the Nobel Assembly at the Karolinska Institute, Sweden for outstanding discoveries in physiology or medicine. The Nobel Prize is presented annually on the anniversary of Alfred Nobel's death, 10 December.
 - As of 2024, 115 Nobel Prizes in Physiology or Medicine have been awarded to 229 laureates, 216 men and 13 women.
 - The first one was awarded in 1901 to the German physiologist, Emil von Behring, for his work on serum therapy and the development of a vaccine against diphtheria.
 - The first woman to receive the Nobel Prize in Physiology or Medicine, Gerty Cori, received it in 1947 for her role in elucidating the metabolism of glucose, important in many aspects of medicine, including treatment of diabetes.
 - The most recent Nobel prize in Physiology or Medicine 2024 was announced by the Karolinska Institute on 7 October, 2024, and has been awarded jointly to American Victor Ambros and Gary Ruvkun, for their discovery of microRNA and its role in post-transcriptional gene regulation.
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- **Electromyography (EMG)** : It is an electrodiagnostic medicine technique for evaluating and recording the electrical activity produced by skeletal muscles. EMG is performed using an instrument called electromyograph to produce a record called electromyogram.
 - **Audiogram** : It is a graph that shows the audible threshold for standardized frequencies as measured by an **audiometer**. The Y-axis represents intensity measured in decibels and the X-axis represents frequency measured in hertz. The threshold of hearing is plotted relative to a standardized curve that represents 'normal' hearing.
 - **Mammography** : Mammography is the process of examining human breast using low-energy X-ray. The actual test is called mammogram. The purpose of mammography screening is **breast cancer detection**.
 - **Endoscope** : It is an illuminated optical, typically slender and tubular instrument (a type of borescope) used to look deep into the body and used in procedures called an endoscopy.
 - **Cardiograph** : It is an instrument that graphically registers movements of the heart.
 - **Crescograph** : A crescograph is a device for measuring the growth in plants. It was invented in the early 20th century by **Jagdish Chandra Bose**.
 - **Tetraplegia** : It is also known as **quadriplegia**. It is a paralysis caused by illness or injury that results in the partial or total loss of use of all four limbs and torso. **Paraplegia** is similar but does not affect the arms. The loss is usually sensory and motor which means that both sensation and control are lost. **Tetraparesis** (quadriparesis), on the other hand, means muscle weakness affecting all four limbs.
 - **Electron Beam Therapy** : It is a kind of external beam radiotherapy where electrons are directed to a tumour site.
 - **Cardiologist** : A cardiologist specializes in diagnosing and treating diseases of the cardiovascular system. The cardiologist will carry out tests, and they may do some procedures, such as heart catheterizations, angioplasty or inserting a pacemaker.
 - **Nephrologist** : The nephrologist deals with the diagnosis and management of kidney diseases. The kidneys are vital for maintaining normal fluid and electrolyte balance in the body.
 - **Urologist** : Urologist treats general diseases of the urinary tract.
 - **Oculist** : An oculist is a person who examines or treats people's eye.
 - **Ultrasound Technique** : Medical ultrasound (also known as diagnostic sonography or ultrasonography) is a diagnostic imaging technique based on the application of the ultrasound. It is used to create an image of internal body structures such as tendons, muscles, joints, blood vessels, and internal organs.
 - **Fetal ultrasound** is a test used during pregnancy that creates an image of the fetus in the mother's womb.
 - **Linus Carl Pauling** : American Linus Carl Pauling has given his important contribution for research in the field of cancer and vitamin-C. He has researched in the field of physics, chemistry and biology. He was awarded Nobel Prize in 1954 (Chemistry) and in 1962 (Peace) for his achievements.
 - **Honey Bees** : These are eusocial flying insect within the genus **Apis** of the bee clade. They are known for construction of perennial, colonial nests from wax, for the large size of their colonies and for their surplus production and storage of honey, distinguishing their hives as a prized foraging target of many animals, including honey badgers, bears and human hunter-gatherers.

- Honey bees are trained to sniff out **landmines**.
- A honey bee colony has one queen, some males (drones) and many workers. We obtain honey and wax from honey bee hives. The average speed of honey bees are normally about 12-15 miles/hour (17 to 21 km/hour). It is noteworthy that the honey bee hive has a **special odour** which helps each other members in identification.
- **Karl Von Frisch** discovered that bees communicate the location of food sources to each other through the complex circle and waggle dances. For this discovery, he was awarded Nobel Prize in 1973.
- **Round Dance** : Round dance is a term used to describe the communicative behaviour of a foraging honey bee, in which she moves on the comb in close circles, alternating right and then left. The round dance indicates that the forger has located a profitable food source close to the hive (approximately 75 m range).
- **Waggle Dance** : The waggle dance tells the watching bees two things about a flower patch's location-the distance and the direction away from the hive. The dancing bee

waggles back and forth as she moves forward in a straight line, then circles around to repeat the dance. It indicates that the food source is far from the hive.

- **Silica gel packets** are used to absorb moisture and keep things (as medicines) dry.
- **Shanti Swarup Bhatnagar Award** : This is a prestigious science award in India given annually by the Council of Scientific and Industrial Research (CSIR) for notable and outstanding research, applied or fundamental, in biology, chemistry, environmental science, engineering, mathematics, medicine and physics. The purpose of this prize is to recognize outstanding Indian work in science and technology. It is the most coveted award in multidisciplinary science in India. This award is named after the Founder Director of the CSIR– Shanti Swarup Bhatnagar. It was first awarded in 1958.
- **Human Eye** : The human eye (sight sense organ) is a spherical body also known as the eyeball. It lies in the eye socket and is attached to various muscles. **Ophthalmology** is a branch of medicine and surgery that deals with the diagnosis and treatment of eye disorders.
- In Eye donation, the **cornea** is the part which is transplanted.
- **Dehydration** : Dehydration is the loss of water or liquid from the body in excessive quantity. In this state the electrolytes mainly sodium chloride is lost from the body.

Different Agricultural Revolutions and Related Fields

Revolutions	Fields
Green Revolution	Cereal production
White Revolution	Milk and Dairy production
Yellow Revolution	Oilseeds production
Blue Revolution	Fisheries and aquaculture
Pink Revolution	Meat and poultry processing/ Onions/Prawn production
Black Revolution	Petroleum production
Silver Revolution	Eggs production
Red Revolution	Meat/Tomato production
Round Revolution	Potato production
Grey Revolution	Fertilizers production
Brown Revolution	Leather production
Golden Revolution	Horticulture/Honey/Fruit pro-duction
Golden Fibre Revolution	Jute production
Silver Fibre Revolution	Cotton production

Institutions and their Locations

Institutions	Location
National Institute of Virology	Pune
National Institute of Immunology	New Delhi
National Centre for Cell Science	Pune
Centre for DNA Fingerprinting and Diagnostics	Hyderabad (Telangana)
National Institute of Plant Genome Research	New Delhi
National Brain Research Centre	Manesar, Gurugram (Haryana)
National Bureau of Animal Genetic Resources	Karnal (Haryana)

Institute of Life Sciences	Bhubaneswar (Odisha)
National Centre for Biological Sciences	Bengaluru (Karnataka)
Birbal Sahni Institute of Palaeosciences	Lucknow (U.P.)
Institute of Bioresources and Sustainable Development	Imphal (Manipur)
National Academy of Medical Sciences	New Delhi
Institute of Forest Genetics and Tree Breeding	Coimbatore (Tamil Nadu)
Institute of Microbial Technology	Chandigarh
Central Institute of Medicinal and Aromatic plants	Lucknow
National Institute for the Empowerment of Persons with Visual Disabilities	Dehradun
National Institute for Locomotor Disabilities	Kolkata
Ali Yavar Jung National Institute of Speech and Hearing Disabilities	Mumbai
National Institute for the Empowerment of Persons with Intellectual Disabilities	Secunderabad (Telangana)
National Botanical Research Institute	Lucknow
National Bureau of Plant Genetic Resources	New Delhi
Indian Institute of Sugarcane Research	Lucknow
National Sugar Institute (Indian Institute of Sugar Technology)	Kanpur (U.P.)
Sugarcane Breeding Institute	Coimbatore (Tamil Nadu)
Indian Veterinary Research Institute	Izatnagar (Bareilly, U.P.)
Central Avian Research Institute	Izatnagar
Central Drug Research Institute	Lucknow
National Rice Research Institute	Cuttack
Central Potato Research Institute	Shimla

Question Bank

1. AYUSH is an acronym for :

- Ayurveda, Yoga and Naturopathy, Unani, Siddha and Homeopathy
- Allopathy, Yoga and Naturopathy, Unani, Siddha and Homeopathy
- Allopathy, Yoga, Unani, Siddha and Homeopathy
- Ayurveda and Allopathy, Yoga, Unani, Siddha and Homeopathy

M.P. P.C.S. (Pre) 2022

Ans. (a)

AYUSH is an acronym for Ayurveda, Yoga and Naturopathy, Unani, Siddha and Homeopathy and are the six Indian systems of medicine prevalent and practiced in India and some of the neighboring Asian countries with very few exceptions in some of the developed countries. Earlier, the Department of Indian System of Medicine and Homoeopathy (ISM & H) formed in 1995, was responsible for the development of these systems. It was then renamed as the Department of Ayurveda, Yoga, and Naturopathy, Unani, Siddha and Homoeopathy (Ayush) in November, 2003 with focused attention towards education and research in Ayurveda, Yoga and Naturopathy, Unani, Siddha and Homoeopathy. Finally, the Ministry of Ayush was formed on the 9th November, 2014 with a vision of reviving the profound knowledge of our ancient systems of medicine and ensuring the optimal development and propagation of the Ayush systems of healthcare.

2. 'Red Ribbon Express' is a visual and moving medium for :

- Awareness about AIDS
- Awareness about Pulse Polio
- Programme for protecting girl child
- Programme for protecting all the children from malnutrition

U.P.P.C.S. (Mains) 2010

Ans. (a)

'Red Ribbon Express' is a visual and moving medium to create and increase awareness about the means of transmission, prevention and perception of risk associated with HIV/AIDS.

3. 'SANRAKSHA' is –

- Malaria Research and Control Project (Bengaluru)
- AIDS Research and Control Project (Bengaluru)
- T.B. Research and Control Project (New Delhi)
- None of these

U.P.P.C.S. (Mains) 2014

Ans. (b)

AIDS program, Sanraksha (Bengaluru) was started as an HIV counselling service in 1993 and today offers a continuum of services that range from HIV testing to hospice care.

4. 'AVAHAN' the Indian segment of 'Bill and Melinda Gates Foundation', is engaged in the prevention of –

(a) Dengue (b) Polio
(c) HIV/AIDS (d) Filariasis

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

Avahan is an initiative sponsored by the Bill & Melinda Gates Foundation to reduce the spread of HIV in India. It began in 2003.

5. NACO is an organisation dealing with :

(a) Cancer (b) Child Healthcare
(c) AIDS (d) None of these

U.P.P.C.S. (Mains) 2006

Ans. (c)

The National AIDS Control Organisation (NACO), established in 1992, is a division of India's Ministry of Health and Family Welfare that provides leadership to HIV/AIDS control programme in India through HIV/AIDS Prevention and Control Societies and is "the nodal organisation for formulation of policy and implementation of programs for prevention and control of HIV/AIDS in India".

6. 'Mission Indradhanush' of the Union Government is related to –

(a) Pulse production
(b) Immunization of children against seven vaccine-preventable diseases
(c) Skill development
(d) Tourism promotion

U.P.R.O./A.R.O. (Pre) 2014

Ans. (b)

Mission Indradhanush was launched by Ministry of Health and Family Welfare (MOHFW), Government of India on 25th December, 2014. The objective of this mission is to ensure that all children under the age of two years as well as a pregnant woman are fully immunized against the seven vaccine preventable diseases.

The mission Indradhanush, depicting seven colours of the rainbow targets to immunize all children against seven vaccines preventable disease, namely : Diphtheria, Pertussis, Tetanus, Tuberculosis, Polio, Hepatitis-B and Measles.

7. The mission 'Indradhanush' launched by the Government of India is in connection with :

(a) Education of children
(b) Immunization of children
(c) Social upliftment
(d) None of the above

U.P.P.S.C. (GIC) 2017

Ans. (b)

See the explanation of above question.

8. Mission Indradhanush Campaign in India is associated with –

(a) Nutrition to pregnant women
(b) Awareness of diabetes
(c) Eradication of blindness
(d) Vaccination of children

U.P.P.C.S. (Mains) 2015

Ans. (d)

See the explanation of above question.

9. Consider the following statements in the context of interventions being undertaken under Anaemia Mukh Bharat Strategy :

1. It provides prophylactic calcium supplementation for pre-school children, adolescents and pregnant women.
2. It runs a campaign for delayed cord clamping at the time of child-birth.
3. It provides for periodic deworming to children and adolescents.
4. It addresses non-nutritional causes of anaemia in endemic pockets with special focus on malaria, haemoglobinopathies and fluorosis.

How many of the statements given above are correct?

(a) Only one (b) Only two
(c) Only three (d) All four

I.A.S. (Pre) 2023

Ans. (c)

Under Anaemia Mukh Bharat (AMB) strategy, the interventions for tackling the problem of anaemia in all the States and UTs include :

- (i) Prophylactic Iron and Folic Acid Supplementation in all six target age groups (including pre-school children, adolescents and pregnant women).
- (ii) Intensified year-round Behaviour Change Communication (BCC) Campaign for : (a) improving compliance to Iron Folic Acid supplementation and deworming, (b) enhancing appropriate infant and young child feeding

practices, (c) encouraging increase in intake of iron-rich food through diet diversity/quantity/frequency and/or fortified foods with focus on harnessing locally available resources, and (d) ensuring delayed cord clamping after delivery (by 3 minutes) in health facilities.

(iii) Testing and treatment of anaemia, using digital methods and point of care treatment, with special focus on pregnant women and school-going adolescents.

(iv) Addressing non-nutritional causes of anaemia in endemic pockets with special focus on malaria, haemoglobinopathies and fluorosis.

Hence, statement 1 is incorrect, while other three statements are correct.

10. By which year has 'National Sickle Cell Anaemia Elimination Mission' fixed the target to eliminate sickle cell anaemia from India?

- (a) 2032 (b) 2037
(c) 2042 (d) 2047

M.P. P.C.S. (Pre) 2025

Ans. (d)

National Sickle Cell Anaemia Elimination Mission (NSCEM) was launched by PM Narendra Modi on 1st July, 2023 in Shahdol, Madhya Pradesh. This mission focuses on addressing the significant health challenges posed by sickle cell disease, particularly among tribal populations of the country. Sickle cell disease (SCD) is a chronic single gene disorder causing a debilitating systemic syndrome characterized by chronic anaemia, acute painful episodes, organ infarction and chronic organ damage and by a significant reduction in life expectancy. NSCEM aims to eliminate sickle cell genetic transmission by the year 2047, showing a long-term commitment to eradicating the disease.

11. Match List-I with List-II and select the correct answer using the codes given below the lists :

List-I (Programme)	List-II (Launch Year)
A. National Programme for Control of Blindness	1. 1975
B. Integrated Child Development Scheme	2. 1976
C. National Leprosy Eradication Programme	3. 2005
D. National Rural Health Mission	4. 1983

Codes :

	A	B	C	D
(a)	1	2	3	4
(b)	2	4	1	3
(c)	2	1	4	3
(d)	4	3	2	1

U.P.P.C.S. (Pre) 2019

Ans. (c)

The correct match of List-I with List-II is as follows :

Programme	Launch Year
National Programme for Control of Blindness	– 1976
Integrated Child Development Scheme	– 1975
National Leprosy Eradication Programme	– 1983
National Rural Health Mission	– 2005

12. National Family Health Survey (NFHS)-5 does not include

- (a) High blood Pressure (b) Diabetes
(c) HIV Test (d) Registration of death

U.P.P.C.S. (Pre) 2019

Ans. (c)

National Family Health Survey (NFHS)-5 did not include HIV test. Pre-school education, registration of deaths and prevalence of diseases like high blood-pressure, diabetes etc. were included in it.

13. Which of the following is not correctly matched?

- (a) Chlorophyll – Magnesium
(b) Haemoglobin – Iron
(c) Table salt – Sodium
(d) Washing soda – Potassium

U.P.R.O./A.R.O. (Mains) 2014

Ans. (d)

The correctly matched order is :

Chlorophyll	–	Magnesium
Haemoglobin	–	Iron
Table salt	–	Sodium chloride
Washing soda	–	Sodium carbonate

14. Match List-I with List-II and select the correct answer from the code given below the lists.

List-I	List-II
A. Acid present in Vinegar	1. Calcium phosphate
B. Souring of milk	2. Hydrochloric acid
C. Compound present in bones	3. Acetic acid
D. Acid present in Gastric juice	4. Lactic acid

Code :

	A	B	C	D
(a)	2	1	4	3
(b)	3	4	1	2
(c)	4	3	1	2
(d)	2	4	1	3

U.P. R.O./A.R.O. (Mains) 2016

Ans. (b)

The correctly matched lists are as follows :

List-I		List-II
Acid present in Vinegar	–	Acetic acid
Souring of milk	–	Lactic acid
Compound present in bones	–	Calcium phosphate
Acid present in Gastric juice	–	Hydrochloric acid

15. Match List-I with List-II and select the correct answer using the code given below the lists :

List-I	List-II
A. Compounds present in bones	1. Lactic acid
B. Acid present in vinegar	2. Ethylene
C. Souring of milk	3. Acetic acid
D. Ripening of fruits	4. Calcium phosphate

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	2	1	4	3
(c)	3	4	2	1
(d)	4	3	1	2

U.P. R.O./A.R.O. (Pre) 2016

Ans. (d)

The correctly matched lists are as follows :

Compounds present in bones	–	Calcium phosphate
Acid present in vinegar	–	Acetic acid
Souring of milk	–	Lactic acid
Ripening of fruits	–	Ethylene

16. Match List-I with List-II and select the correct answer using the codes given below the lists :

List-I	List-II
A. Skin cancer	1. Chloro-fluorocarbons
B. Noise pollution	2. Ultraviolet Rays
C. Global warming	3. Decibel
D. Ozone hole	4. Carbon dioxide

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	2	3	4	1
(c)	2	3	1	4
(d)	4	3	2	1

U.P. Lower Sub. (Pre) 2015

Ans. (b)

The correctly matched lists are as follows :

Skin Cancer	–	Ultraviolet Rays
Noise pollution	–	Decibel
Global warming	–	Carbon dioxide
Ozone hole	–	Chloro-fluorocarbons

17. Which of the following material is bio-degradable?

- (a) Plastic cups (b) Wool
(c) Glass bottle (d) Aluminium foil

U.P. P.C.S. (Pre) 2022

Ans. (b)

One of wool's greatest properties as a sustainable material is its biodegradability. Wool is made out of keratin, the same protein as human hair. Micro-organisms in soil or water can break down woolen matter. Wool's biodegradation properties are in sharp contrast to synthetic fibres, which take much longer to break down and also release microplastics into the environment.

18. Biodegradable wastes can usually be converted into useful substance with the help of :

- (a) bacteria (b) viruses
(c) nuclear proteins (d) radioactive substances
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (a)

Biodegradation is nature's way of recycling wastes, or breaking down organic matter into nutrients that can be used and reused by other organisms. Biodegradable wastes include any organic matter in waste which can be broken down into carbon dioxide, water, methane, compost, humus, simple organic molecules by micro-organisms (comprising mainly bacteria, yeast and fungi) and other living things by composting, aerobic digestion, anaerobic digestion or similar processes. Hence, Biodegradable wastes can usually be converted into useful substance with the help of bacteria.

19. Which among the following is not a biodegradable pollutant?

- (a) Paper
(b) Food and Plant waste
(c) Insecticides
(d) Wood and Cellulose

M.P. P.C.S. (Pre) 2023

Ans. (c)

Among the given options, insecticides are non-biodegradable, while paper, food and plant waste, wood and cellulose are biodegradable materials.

20. Match List-I with List-II and select the correct answer using the codes given below the lists :

List-I	List-II
A. Lactobacillus	1. Alcohol production
B. Rhizobium	2. Citric acid formation
C. Yeast	3. Preparation of curd
D. Aspergillus	4. Nitrogen Fixation

Code :

	A	B	C	D
(a)	4	1	2	3
(b)	1	2	3	4
(c)	3	4	1	2
(d)	4	3	1	2

U.P.P.C.S. (Mains) 2003

Ans. (c)

The correctly matched lists are as follows :

Lactobacillus	–	Preparation of curd
Rhizobium	–	Nitrogen Fixation
Yeast	–	Alcohol production
Aspergillus	–	Citric acid formation

21. Which one of the following type of microorganisms is most widely used in industries ?

- Bacteria
- Bacteria and fungi
- Bacteria and algae
- Bacteria, microalgae and fungi

I.A.S. (Pre) 1998

Ans. (d)

Bacteria, fungi and microalgae, all three are widely used in industries. Today bacteria is used in manufacturing curd, cheese, butter, vinegar and in making of various antibiotic medicines. Diatoms are a small group of algae. Diatomaceous soil consists of fossilized remains of diatoms, a type of hard-shelled algae. It is used in the manufacturing of glass and it is also used in sugarcane mills as bacteria filters. Fungus (yeast) have been used for thousands of years in the production of beer, wine and bread. Besides this different types of acids like citric acid, gallic acid etc. are also obtained from different types of fungus.

22. A group of archaebacteria is used in the production of:

- Ethane
- Methane
- Acids
- Alcohols

R.A.S./R.T.S. (Pre) 1997

Ans. (b)

Methanogens (a group of archaebacteria) are microorganisms that produce methane as a metabolic by-product in hypoxic conditions. They are prokaryotic and belong to the domain of archaea.

23. An organism involved in the fermentation of toddy is :

- Lactobacillus
- Acetobacter
- Saccharomyces
- Penicillium

U.P.P.C.S. (Mains) 2010

Ans. (c)

Saccharomyces is a genus of fungi that include many species of yeasts. It is used in fermentation of toddy (a type of raw alcoholic beverage).

24. Which yeast is used for commercial production of ethanol?

- Candida albicans
- Saccharomyces cerevisiae
- Candida sloofi
- Leucosporidium frigidum

U.P. P.C.S. (Pre) 2016

Ans. (b)

Many yeasts such as the commonly used baker's yeast *Saccharomyces cerevisiae* prefer fermentation to respiration. These yeasts will produce ethanol under aerobic conditions if they are provided with right kind of nutrition.

25. A microbe commonly used for alcoholic fermentation in wine industry is a :

- Yeast
- Bacterium
- Cyanobacterium
- Actinomycete

U.P.P.C.S. (Mains) 2005

Ans. (a)

Yeasts have two main uses in food production :

1. baking, 2. making alcoholic beverages.

The alcohol in wine is formed by the fermentation of the sugars in grape juice, with carbon dioxide as a byproduct. Yeast is commonly used for alcoholic fermentation in the wine industry.

26. Consider the following statements :

- Some microorganisms can grow in environments with temperature above the boiling point of water.
- Some microorganisms can grow in environments with temperature below the freezing point of water.
- Some microorganisms can grow in highly acidic environment with a pH below 3.

How many of the above statements are correct?

- Only one
- Only two
- All three
- None of these

I.A.S. (Pre) 2023

Ans. (c)

Organisms that grow at optimum temperatures of 50°C to a maximum of 80°C are called thermophiles (heat loving). Higher up on the extreme temperature scale we find the hyperthermophiles, which are characterized by growth ranges from 80°C to a maximum of 110°C, with some extreme examples that survive temperatures above 121°C.

Some bacteria grow at the bottom of the ocean in sea vents, where temperatures can reach upto around 700°F. The organisms retrieved from arctic lakes such as Lake Whillans are considered extreme psychrophiles (cold loving). Psychrophiles are microorganisms that can grow at 0°C and below, have an optimum growth temperature close to 15°C, and usually do not survive at temperatures above 20°C. They are found in permanently cold environments such as the deep water of the oceans. Acidophiles are microorganisms that shows optimal growth in highly acidic environments. The extreme acidophiles dwell in environments with a pH value below 3. Hence, all 3 statements are correct.

27. Biomagnification means –

- (a) Fast spreading of cancer cells in the body
- (b) Increasing amount of pesticides in the organisms of successive tropic level
- (c) To see the microscopic parts of the body by microscope
- (d) Sudden increase in the number of the individuals of a species in a specific area

R.A.S./R.T.S.(Pre) 2008

Ans. (b)

Biomagnification, also known as bio-amplification or biological magnification, is the increasing concentration of a substance, such as a toxic chemical (e.g. pesticides), in the tissues of tolerant organisms at successively higher levels in a food chain.

28. Assertion (A) : Herbicides can destroy weeds without harming useful plants.

Reason (R) : Herbicides are absorbed in different rates by weeds and crop plants.

Select the correct answer using the code given below Code :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (c) (A) is true but (R) is false
- (d) (A) is false but (R) is true

U.P.P.C.S. (Mains) 2002

Ans. (a)

Herbicides, also commonly known as weedkillers are used to kill unwanted plants. Herbicides are absorbed in different rates by weed and crop plants. Selective herbicides kill specific targets while leaving the desired crop relatively unharmed. It is clear that (A) and (R) both are correct and (R) is the correct explanation of (A).

29. Generally, Orobanche weed is found in –

- (a) Tobacco field
- (b) Gram field
- (c) Rice field
- (d) Wheat field

U.P.R.O./A.R.O. (Pre) 2014

Ans. (a)

Generally, Orobanche weed is found in a tobacco field. Orobanche weed is a parasitic weed which is also found in tomato and potato fields. It's 150 species are spread over much of the world's temperate and subtropical regions.

30. Information about local flora can be obtained from :

- (a) Herbarium
- (b) Auditorium
- (c) Sanitarium
- (d) Planetarium

R.A.S./R.T.S.(Pre) 2010

Ans. (a)

A herbarium is a collection of preserved plants stored, catalogued and arranged systematically for study by professionals and amateurs from many walks of life. A collection like this is a vital reference when you need to identify a plant and also serves to fix forever the identity of thousands of plant names.

31. ICAR Day is celebrated on :

- (a) 23rd December
- (b) 16th July
- (c) 22nd April
- (d) 23rd November

Jharkhand P.C.S. (Pre) 2023

Ans. (b)

The Indian Council of Agricultural Research (ICAR) celebrates its foundation day every year on 16th July. ICAR was established on 16 July, 1929 as a registered society under the Societies Registration Act, 1860 in pursuance of the report of the Royal Commission on Agriculture. From 2023 it has been decided to observe this day as the 'Foundation and Technology Day' of ICAR. ICAR celebrated its 95th Foundation and Technology Day on 16 July, 2023 at National Agriculture Science Complex, Pusa, New Delhi. The ICAR has its headquarters at New Delhi.

32. Consider the following days and arrange them in chronological order :

1. World Health Day
2. National Maritime Day
3. World Athletics Day
4. World Red Cross Day

Select the correct answer from the codes given below:

- (a) 4, 2, 3, 1
- (b) 2, 1, 3, 4
- (c) 2, 3, 4, 1
- (d) 3, 2, 1, 4

U.P.P.C.S. (Pre) 2024

Ans. (b)

World Health Day	–	7 April
National Maritime Day (India)	–	5 April
World Athletics Day	–	7 May
World Red Cross Day	–	8 May
Hence, the correct chronological order is 2, 1, 3, 4.		

33. Match the following :

- | | |
|-----------------------|---------------------------|
| A. 5 June | 1. Food Day |
| B. 1 December | 2. Environment Day |
| C. 5 September | 3. World AIDS Day |
| D. 16 October | 4. Teachers Day |

Code :

- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 2 | 3 | 4 | 1 |
| (b) | 2 | 3 | 1 | 4 |
| (c) | 3 | 2 | 4 | 1 |
| (d) | 1 | 3 | 2 | 4 |

U.P.P.C.S. (Pre) 1992

Ans. (a)

The correctly matched lists are as follows -
 5 June – World Environment Day
 1 December – World AIDS Day
 5 September – Teachers Day
 16 October – World Food Day

34. Which of the following pairs is not correctly matched?

- | Day | Date |
|-------------------------------|--------------------------|
| (a) World Environment Day | – 5 th June |
| (b) World Earth Day | – 22 nd April |
| (c) World Homeopathy Day | – 12 th March |
| (d) International Day of Yoga | – 21 st June |

Jharkhand P.C.S. (Pre) 2023

Ans. (c)

The World Homeopathy Day is celebrated each year on 10th April to pay tribute to Homeopathy and its contribution to the world of medicine. Other given pairs are correctly matched.

35. Match List- I with List- II and select the correct answer using the code given below :

- | List-I | List-II |
|-----------------------------------|------------------------|
| A. World T.B. Day | 1. December 1 |
| B. World AIDS Day | 2. January 31 |
| C. Anti Leprosy Day | 3. September 16 |
| D. International Ozone Day | 4. March 24 |

Code :

- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 4 | 1 | 2 | 3 |
| (b) | 3 | 4 | 1 | 2 |

- (c) 2 3 4 1
 (d) 1 2 3 4

U.P.P.C.S. (Pre) 1999

Ans. (a)

World T.B. Day	– March 24
World AIDS Day	– December 1
Anti Leprosy Day	– Last Sunday of January (31 January, 1999 was Sunday)
International Ozone Day	– September 16

36. In July 2004, where is the world largest session of HIV/ AIDS is organized in the world ?

- (a) Bangkok (b) Singapore
 (c) Newyork (d) Rome

R.A.S./R.T.S. (Pre) 2008

Ans. (a)

The International AIDS Conference was held in Bangkok, the capital city of Thailand on 16 July, 2004. The main aim of the conference was to publish new guidelines underlining the effectiveness of anti-retroviral drugs to prevent the transmission of HIV from seropositive mothers to their child. 1st December is celebrated every year as 'World AIDS Day'.

37. 1 December, is celebrated every year as –

- (a) World Environment Day
 (b) World Food Day
 (c) World AIDS Day
 (d) World Literacy Day

U.P. Lower Sub. (Pre) 2009

Ans. (c)

See the explanation of above question.

38. 'World AIDS Day' is celebrated on which date?

- (a) 12th January (b) 1st April
 (c) 1st May (d) 1st December

M.P.P.C.S. (Pre) 2012

Ans. (d)

See the explanation of above question.

39. Consider :

- World Alzheimer's day
- World TB day
- Anti Leprosy day
- World AIDS day

What is the correct sequence in which these national and international days are observed in a calendar year ?

- (a) 1, 2, 3, 4 (b) 4, 3, 2, 1

(c) 2, 1, 3, 4

(d) 3, 2, 1, 4

U.P.P.C.S. (Mains) 2002

Ans. (d)

Anti Leprosy day	– last sunday of January
World TB day	– 24 March
World Alzheimer's day	– 21 September
World AIDS day	– 1 December

40. Which of the following is not correctly matched?

List-I

List-II

- | | |
|-----------------------------|-------------|
| (a) International Women Day | 08 March |
| (b) World Water Day | 22 April |
| (c) World Leprosy Day | 30 June |
| (d) World Cancer Day | 04 February |

U.P. P.C.S. (Mains) 2017

Ans. (b) & (c)

The World Water Day is celebrated on 22 March every year and World Leprosy Day is observed on the last Sunday of January each year. Thus, option (b) and (c) both are not correctly matched.

41. Which date is known as 'Diabetes Day'?

- | | |
|--------------------|-------------------|
| (a) 14th February | (b) 14th May |
| (c) 14th September | (d) 14th November |

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

World Diabetes Day (WDD) is celebrated every year on November 14.

42. Match the following pairs correctly :

List-I

List-II

- | | |
|------------------------------|----------------|
| A. World Minority Rights Day | 1. December 1 |
| B. World AIDS Day | 2. November 18 |
| C. World Animal Day | 3. October 16 |
| D. World Food Day | 4. October 4 |

Code :

- | | | | |
|-------|---|---|---|
| A | B | C | D |
| (a) 2 | 1 | 4 | 3 |
| (b) 3 | 2 | 1 | 4 |
| (c) 4 | 3 | 2 | 1 |
| (d) 2 | 3 | 4 | 1 |

U.P.P.C.S. (Pre) 1999

Ans. (*)

World Minority Rights Day	– December 18
World AIDS Day	– December 1
World Animal Day	– October 4
World Food Day	– October 16

43. Match List-I with List-II and select your answer from the codes given below :

List-I

List-II

- | | |
|--|---------------|
| A. The bird that never makes its nest | 1. Lizard |
| B. The animal that shakes off its tail | 2. Chimpanzee |
| C. Largest living creature | 3. Cuckoo |
| D. Most intelligent animal | 4. Blue whale |

Code :

- | | | | |
|-------|---|---|---|
| A | B | C | D |
| (a) 1 | 2 | 3 | 4 |
| (b) 3 | 1 | 4 | 2 |
| (c) 4 | 3 | 2 | 1 |
| (d) 1 | 3 | 2 | 4 |

U.P. Lower Sub. (Spl.) (Pre) 2004

Ans. (b)

The cuckoo never makes its nest. Lizard can give up its tail with the mere touch and tail again regenerate because it has horizontal division in their vertebrals. Chipmpanzee is said to be most wisest animal and blue whale is the largest living creature.

44. Which State Government launched the State-Level Committee for Vulture Conservation (SLCVC), proposed by the national action for the protection of vultures in India?

- | | |
|----------------|---------------|
| (a) Tamil Nadu | (b) Sikkim |
| (c) Assam | (d) Meghalaya |

69th B.P.S.C. (Pre) 2023

Ans. (a)

In October, 2022 Government of Tamil Nadu launched the State-Level Committee for Vulture Conservation (SLCVC) for coordination between various wings of the government to implement the Action Plan for Vulture Conservation (APVC) 2020-2025.

45. Ten captive-bred Asian Giant Tortoises (Manouria emys), one of the largest tortoise species in mainland Asia, were recently released into which National Park of India?

- | |
|-----------------------------------|
| (a) Ntangki National Park |
| (b) Periyar National Park |
| (c) Pench National Park |
| (d) Khangchendzonga National Park |

69th B.P.S.C. (Pre) 2023

Ans. (a)

Ten captive-bred Asian Giant Tortoises (Manouria emys), one of the largest tortoise species in mainland Asia, were released in December, 2022 into a protected area of Ntangki National Park of Nagaland.

46. Consider the following animals of India :

1. Crocodile

2. Elephant

Which of these is/are endangered species ?

- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2003

Ans. (c)

According to the Red List of International Union for Conservation of Nature (IUCN), among the crocodile species found in India, gharial (*Gavialis gangeticus*) is in critically endangered category while Indian elephant is in endangered category.

47. About whom is said 'India's large wild cats', which became extinct in India?

- (a) Cheetah (b) Leopard
(c) Tiger (d) Lion

R.A.S./R.T.S. (Pre) 1992

Ans. (a)

Cheetah also known as the 'hunting leopard', became extinct in India. Government of India initiated a programme to re-introduce cheetahs to India after they became locally extinct more than 70 years ago. In 2022, as a part of 'Project Cheetah', cheetahs flown in from Namibia and South Africa were introduced to Kuno National Park in India.

48. The term 'Sixth mass extinction/sixth extinction' is often mentioned in the news in the context of the discussion of

- (a) Widespread monoculture practices in agriculture and large-scale commercial farming with indiscriminate use of chemicals in many parts of the world that may result in the loss of good native ecosystems.
(b) Fears of a possible collision of a meteorite with the Earth in the near future in the manner it happened 65 million years ago that caused the mass extinction of many species including those of dinosaurs.
(c) Large scale cultivation of genetically modified crops in many parts of the world and promoting their cultivation in other parts of the world which may cause the disappearance of good native crop plants and the loss of food biodiversity.
(d) Mankind's over-exploitation/misuse of natural resources, fragmentation/loss of natural habitats, destruction of ecosystems, pollution and global climate change

I.A.S. (Pre) 2018

Ans. (d)

During the past 500 million years, there were five 'mass extinctions' during which many species rapidly died. Now scientists warn that the sixth mass extinction of life on Earth is unfolding more quickly than feared and humans are the primary cause, who are pursuing a 'biological annihilation' that is coming faster than feared. According to a study published in journal 'Science' in July, 2017 the planet is on the verge of witnessing sixth mass extinction phase. Previous mass extinctions were due to natural climate changes, huge volcanic eruptions or catastrophic meteor strikes. But this one is due to human activities such as deforestation, over-population, pollution, poaching and extreme weather events tied to man-caused global warming, the study said.

49. Which one of the following is not properly matched?

- (a) Gene splicing and Recombinant DNA technology – Genetic Engineering
(b) A diagnostic test to detect the presence or absence of genetic disorders in unborn child – Amniocentesis
(c) A process by which living organisms break down complex matter into simpler constituents – Biodegradation
(d) An inbuilt time keeping the system in all organisms – Biomass

U.P.P.C.S (Pre) 2010

Ans. (d)

Biomass is biological material derived from living organisms. It most often refers to plants or plant-based materials which are specifically called lignocelluloses biomass. As an energy source, biomass can either be used directly via combustion to produce heat or indirectly after converting it to various forms of biofuel. Conversion of biomass to biofuel can be achieved by different methods which are broadly classified into thermal, chemical and biochemical methods. Remaining pairs are correctly matched.

50. Diseases of inner parts of the human body are diagnosed through :

- (a) Cardiograph
(b) Endoscope
(c) Gyroscope
(d) Crescograph

U.P.P.C.S. (Mains) 2014

Ans. (b)

Endoscopy is a nonsurgical procedure used to examine a person's digestive tract. Using an endoscope, a flexible tube with a light and camera attached to it, the doctor can view a picture of digestive tract on a T.V. monitor.

51. An endoscope is a –

- (a) Narrow telescope (b) Type of camera
(c) Simple microscope (d) None of these

U.P.P.S.C. (R.I.) 2014

Ans. (d)

An endoscope is a medical device consisting of a long, thin, flexible (or rigid) tube which has a light source and a video camera. Images of the inside of the patient's body can be seen on a screen. It is not suitable to categorize endoscope as a type of camera.

52. Which of the following types is used by Computed Tomography employed for visualisation of the internal structure of human body ?

- (a) X-rays
(b) Sound waves
(c) Magnetic resonance
(d) Radioisotopes

I.A.S. (Pre) 2007

Ans. (a)

An X-ray computed tomography (X-ray CT) or computerized axial tomography scan (CAT scan), makes use of computer-processed combinations of many X-ray images taken from different angles to produce cross-sectional (tomographic) images (virtual 'slices') of specific areas of a scanned object, allowing the user to see inside the object without cutting.

53. Barium in a suitable form is administered to patients before an X-ray examination of the stomach, because

- (a) Barium allows X-rays to pass through the stomach on account of its transparency to X-rays.
(b) Barium compound, like magnesium sulphate helps in cleaning the stomach before X-ray examination.
(c) Barium is a good absorber of X-rays and this helps the stomach to appear clearly in contrast with the other regions in the picture.
(d) Barium salts are white in colour and this helps the stomach to appear clearly in contrast with other regions in the picture.

I.A.S. (Pre) 1999

Ans. (c)

Barium is a good absorber of X-rays and this helps the stomach to appear clearly in contrast with the other regions in the picture. So its suitable form is administered to patients before an X-ray examination of the stomach. It is used to diagnose abnormalities of the gastrointestinal tract, such as tumours, ulcers, etc.

54. Match List-I with List-II and select the correct answer with the help of code given below the lists :

List-I	List-II
A. EEG	1. Muscle
B. ECG	2. Eyes
C. EOG	3. Brain
D. EMG	4. Heart

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	3	4	2	1
(c)	2	3	4	1
(d)	4	3	1	2

U.P.P.C.S. (Pre) 2009

Ans. (b)

EEG : Electroencephalography is a process related to the brain in which the electrical activities of the brain are recorded.

ECG : Electrocardiography is the process of the recording of heart's electrical activities.

EOG : Electrooculography is the technique of measuring the resting power of retina.

EMG : Electromyography is a test in which the veins of muscles are checked.

55. Brain diseases are diagnosed by :

- (a) E.E.G. (b) E.E.C.
(c) E.M.G. (d) E.K.G.

U.P.P.C.S. (Pre) 1993

Ans. (a)

Many diseases of the human brain can be identified by Electroencephalograph (EEG). Scientist Hunsberger has recorded the waves of brain by putting to electrodes on it. These waves are explained as a graph which is called as E.E.G.

56. E.E.G. records the activity of :

- (a) Heart (b) Brain
(c) Ear (d) Liver

U.P.P.C.S. (Spl.) (Pre) 2008

Ans. (b)

See the explanation of above question.

57. E.E.G. is done to record the working of :

- (a) Heart (b) Lungs
- (c) Brain (d) Kidney

U.P.P.C.S. (Mains) 2014

Ans. (c)

See the explanation of above question.

58. Match List-I with List-II and select the correct answer with the help of code given below the lists :

List-I	List-II
A. Audiogram	1. Heart
B. E.C.G.	2. Brain
C. E.E.G.	3. Ear
D. Mammogram	4. Breast

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	2	1	3	4
(c)	4	3	2	1
(d)	3	1	2	4

U.P.P.C.S. (Mains) 2012

Ans. (d)

Correctly matched order are :

Audiogram	–	Ear
E.C.G.	–	Heart
E.E.G.	–	Brain
Mammogram	–	Breast

59. The Magnetic Resonance Imaging (MRI) is based on the phenomenon of –

- (a) Nuclear magnetic resonance
- (b) Electron spin resonance
- (c) Electron paramagnetic resonance
- (d) Diamagnetism of human tissues

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (a)

MRI is a non-invasive medical test that physicians use to diagnose and treat medical conditions. It uses a powerful magnetic field, radio frequency pulses, and a computer to produce detailed pictures of organs, soft tissues, bone and all other internal body structures visually. The Magnetic Resonance Imaging (MRI) is based on the phenomenon of nuclear magnetic resonance.

60. MRI is a diagnostic tool, which means –

- (a) Magnetic Resonance Index
- (b) Magnetic Resolution Information
- (c) Magnetic Resonance Imaging
- (d) All of the Above

U.P.P.C.S. (Mains) 2008

Ans. (c)

Magnetic resonance imaging (MRI) is a test that uses magnetic field and pulses of radio wave energy to make pictures of organs and structures inside the body.

61. To diagnose the medical conditions, MRI technique is used. Which is not used in MRI?

- (a) Magnetic field (b) X-ray
- (c) Radio wave (d) Gamma ray
- (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (e)

Magnetic resonance imaging (MRI) is a medical imaging technique used in radiology to form pictures of the anatomy and the physiological processes of the body. MRI scanners use strong magnetic fields, magnetic field gradients, and radio waves to generate images of the organs in the body. MRI technique does not involve use of ionizing radiation (Gamma rays, X-rays and the higher ultraviolet part of the electromagnetic spectrum), which distinguishes it from CT and PET scans.

62. What is MRI?

- (a) Magnetic Record of Intestines
- (b) Magnetic Recording of investigations
- (c) Magnetic Resonance Imaging
- (d) Magnetic Resonance in Intestines

U.P.P.C.S. (Pre) 2012

Ans. (c)

See the explanation of above question.

63. BMD test is held for diagnosis of –

- (a) Dengue (b) Malaria
- (c) Osteoporosis (d) AIDS

U.P.P.C.S. (Pre) 2011

Ans. (c)

BMD (Bone Mineral Density) test is used for the diagnosis of osteoporosis. It is a disease of bones in which the chances of fracture increases. In this disease, the bone mineral density decreases and bone microstructure is destroyed. A BMD test measures how much calcium and other types of minerals are in an area of the bone.

64. BMD test is performed for –

- (a) Arthritis (b) Osteoporosis
(c) Osteomalacia (d) None of these

U.P.P.C.S. (Mains) 2014

Ans. (b)

See the explanation of above question.

65. What is the full form of BMD test?

- (a) Bone Marrow Density
(b) Bone Mineral Density
(c) Bone Marrow Deficiency
(d) Bone Marrow Differentiation

U.P.P.C.S. (Mains) 2015

Ans. (b)

See the explanation of above question.

66. What is lithotripsy?

- (a) Art of writing on stones
(b) Breaking of kidney stone through rays
(c) Carbon dating of stones
(d) Trimming stones for household use

M.P.P.C.S. (Pre) 2010

Ans. (b)

Lithotripsy is a medical treatment in which ultrasound shockwaves is used to break up the stones in the kidney, bladder or ureter. After the procedure, the tiny pieces of stones pass out from the body through urine.

67. Electron beam therapy is a kind of radiation therapy in treat –

- (a) Enlarged prostate gland (b) Gall bladder stone
(c) Certain type of cancer (d) Kidney stone

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (c)

Electron beam radiation therapy is used to treat lymphoma- a form of cancer in the skin. There are 2 types of electron beam radiation therapy, these are (1) spot treatment and (2) total skin electron beam therapy.

68. When doctor tells a person that he/she has been affected with tetraplegia, it means that he/she is suffering from paralysis of –

- (a) Right hand only (b) Left hand only
(c) His both legs only (d) Both hands and both legs

U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (d)

The term tetraplegia refers to a condition that causes total or partial paralysis in both hands and both legs. It is caused by accident or diseases. It is also called quadriplegia.

69. In living beings, nutrition, respiration and synthesis are the three major aspects of :

- (a) Growth
(b) Metabolism
(c) Reproduction
(d) Self-perpetuation

Uttarakhand P.C.S. (Mains) 2002

Ans. (b)

Nutrition, respiration and synthesis are the three major aspects of metabolism. Metabolism is a term that is used to describe all chemical reactions involved in maintaining the living state of the cells and the organism. Metabolism can be conveniently divided into two categories: (I) Catabolism- the breakdown of molecules to obtain energy. (II) Anabolism- the synthesis of all compounds needed by the cells.

70. Another name of artificial silk is –

- (a) Rayon (b) Decron
(c) Fibre Glass (d) Nylon

Jharkhand P.C.S. (Pre) 2010

Ans. (a)

Artificial silk is a synthetic fibre which is also known as rayon. Rayon is a manufactured fibre made from natural sources such as wood and agricultural products that are regenerated as cellulose fibre.

71. Which of the following pairs is NOT correctly matched ?

- (a) Oxytocin – Hormone
(b) Aspartame – Synthetic sweetener
(c) Neoprene – Specialized rubber
(d) Rayon – Modified starch

U.P.P.C.S. (Pre) 2019

U.P.P.C.S. (Pre) 2022

Ans. (d)

Rayon is a manufactured fiber made from natural sources such as wood and agricultural products that are regenerated as cellulose fiber. It is not modified starch. Pairs of other options are correctly matched.

72. Which one of the following pairs is not correctly matched?

- (a) Wool – A protein
(b) Rayon – Modified starch
(c) Rubber – A natural polymer
(d) Fullerene – An allotrope of carbon

U.P.P.C.S. (Pre) 2015

U.P. P.C.S. (Pre) 2022

Ans. (b)

Rayon is derived from regenerated cellulose fibre. Other pairs are correctly matched.

73. Silk fibre, hairs, nails and claw are made of –

- (a) Fibrin (b) Elastin
(c) Keratin (d) Collagen

U.P. Lower Sub. (Pre) 2009

Ans. (c)

Silk fibre, hairs, nails and claw are made with keratin proteins.

74. Which of the following substances is NOT a protein?

- (a) Cotton (b) Hair
(c) Spider Silk (d) Hoof

U.P.P.C.S. (Pre) 2019

Ans. (a)

Cotton is a soft, fluffy staple fiber that grows in a boll, or protective case, around the seeds of the cotton plants. This fiber is almost pure cellulose. Hair, spider silk and Hoof are mainly made of protein.

75. Which one of the following plant fibre is obtained from the stem :

- (a) Cotton (b) Coir
(c) Sunn hemp (d) Kapak

U.P.P.C.S. (Mains) 2010

Ans. (c)

Crotalaria juncea, known as brown hemp, Indian hemp, Madras hemp or sunn hemp is a tropical Asian plant of the legume family (Fabaceae). It is generally considered to have originated in India. It is now widely grown throughout the tropics and subtropics as a source of green manure, fodder and lignified fibre obtained from its stem. It bears yellow flowers and elongated alternated leaves.

76. The sperms used for artificial insemination of cattle are stored in :

- (a) Liquid nitrogen (b) Dry ice
(c) Liquid oxygen (d) Liquid ammonia

Uttarakhand P.C.S. (Pre) 2021

Ans. (a)

There are two methods of freezing and storing semen for artificial insemination of cattle : dry ice and alcohol (–100 degrees F) and liquid nitrogen (–320 degrees F). Liquid nitrogen is preferred because there is no evidence of fertility deterioration with age in it. Fertility gradually declines in semen stored in dry ice-alcohol.

77. Bull semen for the purpose of artificial insemination is stored in :

- (a) Ice (b) Liquid Oxygen
(c) Liquid Nitrogen (d) Liquid Carbon Dioxide

U.P.P.C.S. (Mains) 2007

Ans. (c)

See the explanation of above question.

78. Kadaknath is a breed of :

- (a) Bull (b) Goat
(c) Buffalo (d) Fowl

R.A.S./ R.T.S. (Pre) 2021

Ans. (d)

Kadaknath, also called Kali Masi ("fowl having black flesh"), is an Indian breed of chicken. They originated from Dhar and Jabua, Madhya Pradesh. These birds are mostly bred by the rural poor and tribals. There are three varieties : jet black, golden and pencilled. The meat from this breed has a geographical indication (GI Tag) tag that was approved by the Indian government on 30 July 2018.

79. Which of the following is high milk yielding variety of cow ?

- (a) Dorset (b) Holstein
(c) Sahiwal (d) Red Sindhi

U.P.P.C.S. (Mains) 2006

Ans. (b)

Holstein–Friesians are a breed of cattle known as the world's highest-production dairy animals. They are mainly of Netherlands origin. They are farmed for their large dairy production averaging about 25,000 pounds of milk during each lactation cycle.

80. Which of the following is not correctly matched?

Animal		Breed
(a) Buffalo	–	Bhadawari
(b) Cow	–	Sindhi
(c) Goat	–	Jamunapari
(d) Sheep	–	Tharparkar

U.P. R.O./A.R.O. (Pre) 2021

Ans. (d)

Bhadawari is a breed of buffalo while Sindhi and Tharparkar are breeds of cow. Jamunapari is a breed of goat. Hence, pair of option (d) is not correctly matched.

81. The 'Rashtriya Gokul Mission' announced by the Union Government aims at :

- (a) Protecting the indigenous breeds of cows
- (b) Developing high milk yielding cow breeds
- (c) Curbing slaughtering of cattle
- (d) Eliminating foot and mouth disease in cattle

Uttarakhand P.C.S. (Pre) 2012

Ans. (a)

Rashtriya Gokul Mission is a focused project under national programme for bovine breeding and dairy development, with an outlay of Rs 500 crore during the 12th five year plan. The 'Rashtriya Gokul Mission' aims to conserve and develop indigenous breeds of cows in a focused and scientific manner.

82. Match List-I with List-II and select the correct answer by using the code given below :

- | List-I | List-II |
|----------------|---------------------|
| A. Myoglobin | 1. Radiotherapy |
| B. Sarp Gandha | 2. Oxygen transport |
| C. Carcinoma | 3. Tranquilizer |
| D. Haemoglobin | 4. Muscle cell |

Code :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 3 | 2 | 1 | 4 |
| (b) | 4 | 1 | 2 | 3 |
| (c) | 4 | 3 | 1 | 2 |
| (d) | 2 | 3 | 1 | 4 |

U.P.P.C.S. (Spl) (Mains) 2004

Ans. (c)

The correctly matched lists are as follows :

- | | | |
|-------------|---|------------------|
| Myoglobin | – | Muscle cell |
| Sarp Gandha | – | Tranquilizer |
| Carcinoma | – | Radiotherapy |
| Haemoglobin | – | Oxygen transport |

83. Match List-I with List-II and select the correct answer from the code given below the lists :

- | List-I | List-II |
|----------------|--------------------|
| A. Bulimia | 1. Alkaloid |
| B. Cholesterol | 2. Egg-Yolk |
| C. Atropine | 3. Pancreas |
| D. Insulin | 4. Eating disorder |

Code :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 4 | 2 | 1 | 3 |
| (b) | 3 | 1 | 4 | 2 |

- | | | | | |
|-----|---|---|---|---|
| (c) | 1 | 2 | 3 | 4 |
| (d) | 2 | 4 | 1 | 3 |

U.P.P.C.S. (Spl.) (Mains) 2004

Ans. (a)

The correctly matched lists are as follows :

- | | | |
|-------------|---|-----------------|
| Bulimia | – | Eating disorder |
| Cholesterol | – | Egg-Yolk |
| Atropine | – | Alkaloid |
| Insulin | – | Pancreas |

84. An electronic diode using a semi-conductor nanoparticle material synthesised by microorganisms has been recently fabricated by Indian scientists for the first time, having utility in device miniaturization. The organism used is a/an –

- (a) Virus
- (b) Bacteria
- (c) Yeast
- (d) Protozoa

U.P.P.C.S. (Mains) 2002

Ans. (c)

An electronic diode using a semiconductor nanoparticle material (cadmium sulphide) synthesised by microorganisms has been recently fabricated by Indian scientists for the first time with the help of yeast *Schizosaccharomyces pombe* having utility in device miniaturization. The diode was created by using this cadmium sulphide with phenylene vinylene.

85. The substances which can be used as anaesthetic are –

- 1. Chloroform
- 2. Nitrous oxide
- 3. Helium
- 4. Carbon dioxide

Select the correct answer using the codes given below :

Code :

- (a) 1 and 2
- (b) 2 and 3
- (c) 3 and 4
- (d) 1 and 4

U.P.P.C.S. (Mains) 2002

Ans. (a)

Anaesthetic substances are mainly used for decreasing the sensation or awareness. William Morton has used it for the first time in 1846 in the form of di-ethyl ether. In 1847, Jams Sampson used chloroform as an anaesthetic. Chloroform, Nitrous oxide, pentothal sodium, halothane, chloropropane, cocaine etc. are used as an anesthetic.

86. The gas used as a mild anaesthetic in hospitals is :

- (a) Carbon dioxide
- (b) Carbon monoxide
- (c) Helium
- (d) Nitrous oxide

U.P.P.C.S. (Spl) (Mains) 2008

Ans. (d)

See the explanation of above question.

87. Match List-I with List-II and select the correct answer using the codes given below :

List-I (Specialist)	List-II (Body Part)
A. Cardiologist	1. Eye
B. Nephrologist	2. Urinary tract
C. Urologist	3. Heart
D. Oculist	4. Kidney

Code :

	A	B	C	D
(a)	3	4	2	1
(b)	4	3	1	2
(c)	1	2	4	3
(d)	2	1	3	4

U.P.P.S.C. (GIC) 2010

Ans. (a)

Specialist	Body Part
Cardiologist	– Heart
Nephrologist	– Kidney
Urologist	– Urinary tract
Oculist	– Eye

88. Which one of the following pairs is correctly matched?

(Branch of Medicine)	(Human Organs)
(a) Cardiology	– Respiratory System
(b) Gastroenterology	– Urinary System
(c) Urology	– Digestive System
(d) Neurology	– Nervous System

U.P. Lower Sub. (Mains) 2015

Ans. (d)

Neurology is a branch of medicine dealing with the disorders of the nervous system. Cardiology is a branch of medicine dealing with the disorders of the heart as well as parts of the circulatory system. Gastroenterology is the branch of medicine focused on the digestive system and its disorders. Urology is the branch of medicine that focuses on surgical and medical diseases of the male and female urinary tract system and male reproductive organs.

89. Which one of the following is not banned for use as performance enhancing substance by the athletes :

- | | |
|--------------------------|------------------|
| (a) Human growth hormone | (b) Testosterone |
| (c) Erythropoietin | (d) Colostrum |

U.P.P.C.S. (Mains) 2003

Ans. (d)

Colostrum has been marked as a safe dietary supplement by the International Olympic Committee and it can be used as performance enhancing substance by the athletes. The other options are banned for use by the athletes.

90. Which one of the following pairs is not correctly matched?

Compound	Classification
(a) Saccharin	– Carbohydrate
(b) Adrenaline	– Hormone
(c) Thiamine	– Vitamin
(d) Keratin	– Protein

U.P.P.S.C. (GIC) 2017

Ans. (a)

Saccharin is a white crystalline, slightly water-soluble powder, produced synthetically, which is used as an artificial sweetener. It is about 300-400 times as sweet as sucrose. It is unrelated to the carbohydrates. Pairs of other options are correctly matched.

91. Which one of the following pairs is not correctly matched?

- | | |
|----------------------------|--------------------|
| (a) Control centre of cell | – Nucleus |
| (b) Natural Rubber | – Poly-isoprene |
| (c) Cryolite | – Ore of Aluminium |
| (d) Antiseptic | – Phenolphthalein |

U.P.P.C.S. (Mains) 2003

Ans. (d)

Phenolphthalein is often used as an indicator in acid-base titrations. It is also used as laxative, not as antiseptic. So option (d) is not correctly matched.

92. Which one of the following nonsteroidal contraceptive pills has been developed by CDRI, Lucknow?

- | | |
|-------------|-------------|
| (a) Saheli | (b) Paheli |
| (c) Sangini | (d) Kilhari |

U.P. P.C.S. (Mains) 2017

Ans. (a)

Central Drug Research Institute (CDRI), Lucknow has developed the molecule centchroman (ormeloxifene) which is marketed by HLL under the brand name Saheli. Introduced in 1991, Saheli had the unique distinction of being the world's first non-steroidal oral contraceptive pill. It is also unique since it is an indigenously developed drug. It is once per week pill.

93. Recently developed technique for monitoring foetal growth is :

- | | |
|-------------------------|-----------------|
| (a) Microwave technique | (b) Ultrasonics |
|-------------------------|-----------------|

- (c) Ultraviolet rays (d) X-rays

U.P.P.C.S. (Mains) 2006

Ans. (b)

Ultrasonics is a technique for monitoring foetal growth. It is the application of ultrasound. Ultrasound can be used for medical imaging, detection and measurement. At higher power levels, ultrasonics is useful for changing the chemical properties of substances.

94. Match List-I with List-II and select the correct answer from the code given below the lists :

List-I (Name of Scientist)	List-II (Area of work)
A. Birbal Sahni	1. Green Revolution
B. K.C. Mehta	2. Embryology
C. P. Maheshwari	3. Paleobotany
D. M.S. Swaminathan	4. Plant pathology

Code :

A	B	C	D
(a) 1	2	3	4
(b) 3	4	2	1
(c) 2	1	3	4
(d) 4	3	2	1

U.P. R.O./A.R.O. (Mains) 2021

Ans. (b)

The correctly matched lists are as follows :

Name of Scientist	Area of work
Birbal Sahni	– Paleobotany
K.C. Mehta	– Plant Pathology
P. Maheshwari	– Embryology
M.S. Swaminathan	– Green Revolution

95. Which among these is correctly matched –

- (a) Dr. Kurien – Space
 (b) Dr. Malcom Adiseshiah – Co-operative Movement
 (c) Dr. Abdul Kalam – Development Economics
 (d) Dr. Promod Karan Sethi – Jaipur Prosthetic Foot

I.A.S. (Pre) 1993

Ans. (d)

The correctly matched lists are as follows :

Dr. Verghese Kurien	– Co-operative Movement
Dr. Malcom Adiseshiah	– Development Economics
Dr. Abdul Kalam	– Space
Dr. Pramod Karan Sethi	– Jaipur Prosthetic Foot

96. 'Invasive Species Specialist Group' (that develops Global Invasive Species Database) belongs to which one of the following organizations?

- (a) The International Union for Conservation of Nature
 (b) The United Nations Environment Programme
 (c) The United Nations World Commission for Environment and Development
 (d) The World Wide Fund for Nature

I.A.S. (Pre) 2023

Ans. (a)

The Global Invasive Species Database (GISD) is a free, online searchable source of information about alien and invasive species that negatively impact biodiversity. GISD is managed by the Invasive Species Specialist Group (ISSG) and it was developed between 1998 and 2000 as part of the global initiative on invasive species led by the erstwhile Global Invasive Species Programme (GISP). The Invasive Species Specialist Group (ISSG) is a global network of scientific and policy experts on invasive species, organized under the auspices of the Species Survival Commission (SSC) of the International Union for Conservation of Nature (IUCN).

97. Indian Veterinary Research Institute is located at :

- (a) Bareilly (b) Mathura
 (c) Karnal (d) Patna

U.P. P.C.S. (Pre) 2016

Ans. (a)

Indian Veterinary Research Institute or IVRI is located at Izatnagar, Bareilly in Uttar Pradesh. It is India's premier advanced research facility in the field of veterinary medicine and allied branches.

98. 'Indian Veterinary Research Institute' is located at :

- (a) Bareilly (b) Kanpur
 (c) Aligarh (d) Jhansi

U.P.P.C.S. (Mains) 2013

Ans. (a)

See the explanation of above question.

99. CDRI (India) is located at :

- (a) Allahabad (b) Delhi
 (c) Bangalore (d) Lucknow

R.A.S./R.T.S.(Pre) 1999

U.P.P.C.S. (Pre) 1991

Ans. (d)

CDRI (Central Drug Research Institute) is located at Lucknow.

100. Survey of India is under the ministry of :

- (a) Defence
- (b) Environment and Forests
- (c) Home Affairs
- (d) Science and Technology

I.A.S. (Pre) 2003

Ans. (d)

The Survey of India is India's National Mapping Agency (NMA) in charge of mapping and surveying. It was set up in 1767 and has evolved rich traditions over the years. Survey of India functions under the Ministry of Science and Technology.

101. The headquarters of Survey of India is located in –

- (a) Chandigarh
- (b) Hyderabad
- (c) Dehradun
- (d) New Delhi

Uttarakhand P.C.S. (Pre) 2006

Ans. (c)

The headquarter of Survey of India is located in Dehradun.

102. Which one of the following is an unmatched pair ?

- (a) Central Drug Research Institute - Lucknow
- (b) Central Electrochemical Research Institute - Kolkata
- (c) Centre for Cellular and Molecular Biology - Hyderabad
- (d) Central Salt Research Institute - Bhavnagar

U.P.P.C.S. (Mains) 2004

Ans. (b)

Central Electrochemical Research Institute is one of a chain of forty national laboratories under the aegis of the Council of Scientific and Industrial Research (CSIR). It is situated at Karaikudi in Tamil Nadu. Other pairs are correctly matched.

103. ICAR's Central Institute for Research on Goats is established at :

- (a) Bharatpur
- (b) Makhdoom, Mathura
- (c) Bikaner
- (d) Aligarh

U.P.P.C.S. (GIC) 2017

Ans. (b)

The Central Institute for Research on Goats (CIRG) was established by the Indian Council of Agricultural Research (ICAR), at Makhdoom village near Farah town in Mathura district of Uttar Pradesh on 12 July, 1979 for undertaking research on all aspects of goat production and utilization.

104. Where is the High Security Animal Disease Laboratory is located in India ?

- (a) Hyderabad
- (b) Bangalore

(c) Bhopal

(d) Mumbai

U.P.P.C.S. (Mains) 2005

Ans. (c)

High Security Animal Disease Laboratory is located at Bhopal, Madhya Pradesh.

105. Central Institute of Medicinal and Aromatic Plants is situated at which of the following place?

- (a) Lucknow
- (b) Varanasi
- (c) Gorakhpur
- (d) None of the above

U.P. R.O./A.R.O. (Mains) 2021

Ans. (a)

Central Institute of Medicinal and Aromatic Plants (CIMAP) is a premier multidisciplinary research institute of Council of Scientific and Industrial Research (CSIR) with its major focus on exploiting the potential of medicinal and aromatic plants (MAPs) by cultivation, bioprospection, chemical characterization, extraction, and formulation of bioactive phytochemicals. It is situated at Lucknow. Initially set up as Central Indian Medicinal Plants Organisation (CIMPO) in the year 1957 with a mandate to work and stimulate research on medicinal plants; subsequently aromatic plants also brought under its ambit. CIMPO started functioning from 26th March, 1959 and rechristened as Central Institute of Medicinal and Aromatic Plants (CIMAP) in the year 1978.

106. Among Lucknow based National Science Institutes, identify the non C.S.I.R. Institute :

- (a) CIMAP (Central Institute of Medicinal and Aromatic Plants)
- (b) NBRI (National Botanical Research Institute)
- (c) BSIP (Birbal Sahni Institute of Palaeobotany)
- (d) CDRI (Central Drug Research Institute)

U.P.P.C.S. (Mains) 2005

Ans. (c)

The Birbal Sahni Institute of Palaeobotany (BSIP; now renamed as Birbal Sahni Institute of Palaeosciences) is an autonomous institute or a non-C.S.I.R. Institute constituted under the Department of Science and Technology, Government of India. This Institute is located at Lucknow, Uttar Pradesh and is a place of higher learning in the field of plant fossil research.

107. Which one of the following is not correctly matched?

- | (Institute) | (Location) |
|---|-------------|
| (a) National Centre of Organic Farming | – Ghaziabad |
| (b) Indian Institute of Pulses Research | – Kanpur |

- (c) Indian Institute of Vegetable Research – Gorakhpur
(d) Indian Institute of Sugarcane Research – Lucknow

U.P. R.O./A.R.O. (Mains) 2021

Ans. (c)

Indian Institute of Vegetable Research is situated at Varanasi. Other given Institutes and their locations are correctly matched.

108. The National Institute for the Mentally Handicapped is situated at –

- (a) Hyderabad (b) New Delhi
(c) Kolkata (d) Chennai

U.P.P.C.S. (Pre) 2013

Ans. (a)

The National Institute for the Mentally Handicapped is situated at Secunderabad (District-Hyderabad). It is now renamed as National Institute for the Empowerment of Persons with Intellectual Disabilities.

109. Match List-I with List-II and select the correct answer using the code given below in the lists :

- | List-I
(Institution) | List-II
(Location) |
|--|-------------------------------|
| A. Central Drug Research Institute | 1. Allahabad |
| B. Central Leprosy Institute | 2. Kanpur |
| C. Motilal Nehru National Institute of Technology | 3. Agra |
| D. Indian Institute of Sugar Technology | 4. Lucknow |

Code :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 1 | 2 | 3 | 4 |
| (b) | 3 | 1 | 4 | 2 |
| (c) | 4 | 3 | 1 | 2 |
| (d) | 2 | 4 | 3 | 1 |

U.P.P.C.S. (Pre) 2001

Ans. (c)

The correctly matched order is :
Central Drug Research Institute – Lucknow
National JALMA Institute for Leprosy and Other Mycobacterial Diseases (Central Leprosy Institute) – Agra
Motilal Nehru National Institute of Technology – Allahabad
Indian Institute of Sugar Technology (National Sugar Institute) – Kanpur

110. Match List-I with List-II and select the correct answer using the codes given below in the lists.

- | List-I (Institute) | List-II (Location) |
|--|---------------------------|
| A. Central Institute of Medicinal and Aromatic Plants | 1. Chandigarh |
| B. Centre for DNA Fingerprinting and Diagnostics | 2. Hyderabad |
| C. Institute of Microbial Technology | 3. New Delhi |
| D. National Institute of Immunology | 4. Lucknow |

Code :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 2 | 4 | 1 | 3 |
| (b) | 4 | 2 | 1 | 3 |
| (c) | 2 | 4 | 3 | 1 |
| (d) | 4 | 2 | 3 | 1 |

I.A.S. (Pre) 2001

Ans. (b)

Central Institute of Medicinal and Aromatic Plants – Lucknow
Centre for DNA Fingerprinting and Diagnostics – Hyderabad
Institute of Microbial Technology – Chandigarh
National Institute of Immunology – New Delhi.

111. Where is the National Institute of Virology situated?

- (a) Pune (b) Hyderabad
(c) Mumbai (d) Lucknow

M.P. P.C.S. (Pre) 2020

Ans. (a)

The National Institute of Virology is situated in Pune, Maharashtra. It is part of the Indian Council of Medical Research (ICMR). It was previously known as 'Virus Research Centre' and was founded in collaboration with the Rockefeller Foundation. It has been designated as a WHO H5 reference laboratory for SE Asia region.

112. DNA Fingerprinting and Diagnostics Research Centre is situated at :

- (a) Pune (b) New Delhi
(c) Hyderabad (d) Kolkata

U.P. Lower Sub. (Pre) 2013

Ans. (c)

The Centre for DNA Fingerprinting and Diagnostics (CDFD) is an Indian biotechnology research centre, located in Hyderabad, India, operated by the Department of Biotechnology, Ministry of Science and Technology, Government of India.

113. In which of the following places is Indian Vegetable Research Institute situated?

- (a) Agra (b) Kanpur
(c) Lucknow (d) Varanasi

U.P. R.O./A.R.O. (Mains) 2017

Ans. (d)

Indian Vegetable Research Institute (IVRI) is situated at Varanasi. It is a field unit of Indian Council of Agricultural Research (ICAR).

114. Which of the following pairs is not correctly matched?

- | Institute | Location |
|---|------------|
| (a) Indian Vegetable Research Institute | – Varanasi |
| (b) Central Institute of Subtropical Horticulture | – Srinagar |
| (c) Indian Institute of Sugarcane Research | – Lucknow |
| (d) Indian Institute of Pulse Research | – Kanpur |

U.P. R.O./A.R.O. (Pre) 2017

Ans. (b)

The Central Institute of Subtropical Horticulture (CISH) is situated in Lucknow. It was started as Central Mango Research Station in 1972 under the aegis of the Indian Institute of Horticulture Research, Bengaluru. Other options are correctly matched.

115. Where in Uttar Pradesh is Artificial Limbs Manufacturing Corporation situated?

- (a) Varanasi (b) Prayagraj
(c) Kanpur (d) Agra

U.P. R.O./A.R.O. (Mains) 2017

Ans. (c)

Head Office of Artificial Limbs Manufacturing Corporation of India (ALMICO) is situated in Kanpur, Uttar Pradesh. It was incorporated in 1972 and started manufacturing artificial aids and appliances in 1976.

116. Global Gene Bank of rice is situated at :

- (a) Philippines
(b) Pusa, New Delhi
(c) Cuttack
(d) Hyderabad

Jharkhand P.C.S. (Pre) 2023

Ans. (a)

The International Rice Genebank is situated at the International Rice Research Institute (IRRI), which is headquartered in Los Banos, Philippines. The genebank is the biggest collection of rice genetic diversity in the world. It conserves the largest and most diverse collection of rice germplasm. The facilities of the genebank ensure the long-term preservation of this important diversity. Countries from all over the world send their rice to IRRI for safe keeping, and for sharing for the common public good.

117. Headquarters of International Potato Centre (CIP) is situated in :

- (a) Lima, Peru (b) Shimla, India
(c) Dakar, Senegal (d) Dublin, Ireland

U.P. R.O./A.R.O. (Pre) 2021

Ans. (a)

The International Potato Centre (CIP) was founded in 1971 as a research-for-development organization with a focus on potato, sweetpotato and Andean roots and tubers. It delivers innovative science-based solutions to enhance access to affordable nutritious food, foster inclusive sustainable business and employment growth, and drive the climate resilience of root and tuber agri-food systems. Headquartered in Lima, Peru, CIP has a research presence in more than 20 countries in Africa, Asia and Latin America.

118. Match List-I with List-II and select the correct answer from the code given below the lists :

- | List-I | List-II |
|--------------------------------------|------------|
| A. F.A.O. Headquarters | 1. Shimla |
| B. Central Agmark Laboratory | 2. Cuttack |
| C. Central Rice Research Institute | 3. Rome |
| D. Central Potato Research Institute | 4. Nagpur |

Code :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 1 | 2 | 3 | 4 |
| (b) | 3 | 4 | 2 | 1 |
| (c) | 4 | 3 | 1 | 2 |
| (d) | 3 | 4 | 1 | 2 |

U.P.P.C.S. (Mains) 2009

Ans. (b)

The correctly matched lists are as follows :

- | | | |
|-----------------------------------|---|---------|
| F.A.O. Headquarters | – | Rome |
| Central Agmark Laboratory | – | Nagpur |
| Central Rice Research Institute | – | Cuttack |
| Central Potato Research Institute | – | Shimla |

119. Identify a mismatch among the following pairs :

- (a) NBRI–Kolkata (b) IISR–Lucknow
(c) NPL–New Delhi (d) IISc–Bangalore

U.P.P.C.S. (Mains) 2005

Ans. (a)

The National Botanical Research Institute (NBRI) is a research institute of CSIR in Lucknow. It is engaged in the field of taxonomy and modern biology. Remaining options are correctly matched.

120. National Bureau of Plant Genetic Resources is located at–

- (a) Hyderabad (b) Bangalore
(c) New Delhi (d) Srinagar

U.P.P.C.S. (Mains) 2013

Ans. (c)

The National Bureau of Plant Genetic Resources (NBPGR) has its headquarters at New Delhi.

121. Rajiv Gandhi Centre for Biotechnology is situated at :

- (a) Thiruvananthapuram (b) Faridabad
(c) Jodhpur (d) New Delhi

Rajasthan P.C.S. (Pre) 2023

Ans. (a)

Rajiv Gandhi Centre for Biotechnology is a research institute, exclusive devoted to research in Molecular Biology and Biotechnology. It is located at Thiruvananthapuram, the capital city of the state of Kerala.

122. World's first camel hospital is located in which city?

- (a) Tehran (b) Jaipur
(c) Bikaner (d) Dubai
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (d)

World's first camel hospital is located in Dubai, United Arab Emirates. Its name is 'Dubai Camel Hospital', that exclusively treats camels.

123. Camel uses its hump for :

- (a) storing of water
(b) storing of fat
(c) balancing the body during walking in desert sand
(d) temperature regulation

I.A.S. (Pre) 1993

Ans. (b)

There is about 50 kg huge hump on the camel's spine. This is for storing fat. Camels use this fat as an energy source during their long journey. It is notable that camel is said as 'ship of the desert' because of its favourable features. They also have padded legs and can drink gallons of water at a time.

124. Who declared the year 2024 as International year of Camelids?

- (a) UN (b) UNESCO
(c) WHO (d) USA

M.P. P.C.S. (Pre) 2023

Ans. (a)

United Nations has designated 2024 as the International Year of Camelids, recognizing their pivotal role in the livelihoods of millions, especially among pastoralists in arid and mountainous regions.

125. Match List-I with List-II and select the correct answer from the code given below the lists :

List-I

List-II

- | | |
|------------------------|----------------------------------|
| A. Testosterone | 1. Sedative drug |
| B. Codeine | 2. Rubbersource substance |
| C. Caoutchouc | 3. Aromatic oil of clove |
| D. Eugenol | 4. Hormone |

Codes :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 4 | 1 | 2 | 3 |
| (b) | 1 | 2 | 3 | 4 |
| (c) | 4 | 3 | 2 | 1 |
| (d) | 2 | 3 | 4 | 1 |

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (a)

The correctly matched lists are as follows :

Testosterone	–	Hormone
Codeine	–	Sedative drug
Caoutchouc	–	Rubber source substance
Eugenol	–	Aromatic oil of clove

126. Match List-I with List-II and select the correct answer using the code given below the lists :

List-I

List-II

- | | |
|-----------------------|---|
| A. Anthrax | 1. Disease due to a defective gene |
| B. Thalassemia | 2. Womb-renting |
| C. Surrogacy | 3. Science of altering genomes |
| D. Transgenics | 4. A toxin used in biowarfare |

Code :

	A	B	C	D
(a)	4	1	2	3
(b)	2	3	4	1
(c)	3	2	1	4
(d)	1	2	3	4

U.P.P.C.S. (Mains) 2003

Ans. (a)

The correctly matched lists are as follows :

Anthrax	–	A toxin used in biowarfare
Thalassemia	–	Disease due to a defective gene
Surrogacy	–	Womb-renting
Transgenics	–	A science of altering genomes

127. Which one of the following is not correctly matched ?

- | | | |
|-------------------------|---|-------------------|
| (a) Antifreeze compound | – | Ethylene glycol |
| (b) Antiknock agent | – | Tetraethyl lead |
| (c) Antioxidant | – | β -Carotene |
| (d) Antibiotics | – | Quinine |

U.P.P.C.S. (Mains) 2010

Ans. (d)

Quinine is a white crystalline alkaloid matter. It is used to treat malaria, fever, chills, body aches and swelling. It is not an antibiotic. Remaining pairs are correctly matched.

128. In which Indian city the first dianosaurium has been opened ?

- | | |
|---------------|---------------|
| (a) Mumbai | (b) Jalandhar |
| (c) Hyderabad | (d) Ahmedabad |

U.P.P.C.S. (Mains) 2004

Ans. (c)

The first dianosaurium has been opened in Hyderabad. This dianosaurium has been opened in B.M Birla science center in Hyderabad.

129. Headquarters of 'World Health Organization' -

- | | |
|--------------|------------|
| (a) Geneva | (b) Paris |
| (c) New York | (d) Berlin |

M.P.P.C.S. (Pre) 1995

Ans. (a)

WHO (World Health Organization) headquarter is located in Geneva.

130. Where was the first Dental College of the country founded?

- | | |
|---------------------|------------------------|
| (a) Aligarh | (b) Calcutta (Kolkata) |
| (c) Bombay (Mumbai) | (d) Madras (Chennai) |

M.P.P.C.S. (Pre) 2004

Ans. (b)

The first Dental College of the country was founded in 1920 in Kolkata. Its name was R. Ahmed Dental College.

131. "By Weaving threads of physics, chemistry and biology into a rich tapestry, this remarkable scientist provided a unifying molecular view of the world. He touted the wonderful properties of Vitamin C first as a cure of common cold and later as preventing agent against cancer". One of the science's major figures of all time referred to above is :

- | | |
|------------------|------------------------|
| (a) G..N. Lewis | (b) Linus Carl Pauling |
| (c) Fritz London | (d) Walter Heitle |

I.A.S. (Pre) 1995

Ans. (b)

Linus Carl Pauling contributed to the research work in the field of cancer & ascorbic acid (Vitamin C). He has done a huge research in the field of physics, chemistry and biology. In 1954 he was awarded the Nobel Prize in chemistry. In 1962 he was awarded the Nobel Peace Prize for his opposition to weapons of mass destruction.

132. Which one of the following are useful in the detection of landmines :

- | | |
|---------------|----------|
| (a) Bee | (b) Wasp |
| (c) Butterfly | (d) Moth |

U.P.P.C.S. (Pre) 2002

Ans. (a)

The honey bee is a social insect which is useful in the detection of landmines. A honey bee is a member of the genus Apis, primarily distinguished by the production and storage of honey and the construction of perennial, colonial nests from wax.

133. What is the average speed of a bee?

- | | |
|-------------|-------------|
| (a) 10 km/h | (b) 5 km/h |
| (c) 1 km/h | (d) 16 km/h |

M.P.P.C.S. (Pre) 2010

Ans. (d)

Bees are the member of genus Apis. They fly on the average speed of about 15 miles/hour (21 kmph) while their speed in returning with food is approx. 12 miles/hour (17 kmph). So the closest answer is (d).

134. How do the members of honey-bee hive recognise each other?

- (a) By smell (b) By eyes
(c) By dancing (d) By touching

39th B.P.S.C. (Pre) 1994

Ans. (a)

The members of honey-bee hive recognise each other by smell. The honey bee hive has a special odour which helps each other members in identification. Bees communicate the location of food sources to each other through the complex circle and waggle dances.

135. Which of the following organisms perform waggle dance for others of their kin to indicate the direction and the distance to a source of their food?

- (a) Butterflies (b) Dragonflies
(c) Honeybees (d) Wasps

I.A.S. (Pre) 2023

Ans. (c)

Waggle dance is a term used in beekeeping and ethology for a particular figure-eight dance of the honey bee. By performing this dance, successful foragers can share information about the direction and distance to a good source of food i.e. patches of flowers yielding nectar and pollen, to water sources, or to new nest-site locations with other members of the colony.

136. The Nobel Prize for deciphering the language of bees was awarded to –

- (a) H.G. Khurana (b) K.V. Frisch
(c) Julian Huxley (d) Dorothy Hodgkins

I.A.S. (Pre) 1995

Ans. (b)

German-Austrian ethologist Karl Ritter von Frisch got Nobel Prize in physiology or medicine in 1973 for his work on the investigations of the sensory perceptions and deciphering the language of the honey bee. He was one of the first to translate the meaning of the waggle dance of honey bee.

137. Which one is a stimulant?

- (a) Alcohol (b) Marijuana
(c) Cigarette (d) Opium

Chhattisgarh P.C.S. (Pre) 2003

Ans. (c)

Alcohol, marijuana and opium are depressant substances, while cigarette smoke acts as a stimulant.

138. A small pouch containing silica gel is often found in bottles of medicine in tablet or powder form because of silica gel :

- (a) kills bacteria
(b) kills germs and spores
(c) absorbs moisture
(d) absorbs all gases present inside the bottle

I.A.S. (Pre) 2000

Ans. (c)

Silica gel is most commonly encountered in everyday life as beads in a small (typically 2 x 3 cm) paper packet. In this form, it is used as a desiccant to control local humidity to avoid spoilage or degradation of packed goods. Because silica gel can have added chemical indicators and absorbs moisture very well, silica gel packets usually bear warnings for the user not to eat the contents. It is mainly used for the packed medicines.

139. For outstanding contribution to which one of the following fields is Shanti Swarup Bhatnagar Prize given?

- (a) Literature (b) Performing Arts
(c) Science (d) Social Service

I.A.S. (Pre) 2009

Ans. (c)

Shanti Swarup Bhatnagar was a well-known Indian scientist, a professor of chemistry and scientific administrator. He was the Founder Director and later the first Director-General of Council of Scientific and Industrial Research (CSIR), and he is revered as the 'father of research laboratories'. He was also the first Chairman of the University Grants Commission (UGC). To honour his name and achievements, CSIR instituted an award Shanti Swarup Bhatnagar Prize for Science and Technology, since 1958 for outstanding scientists who made significant contributions in various branches of science.

140. In eye donation, which one of the following parts of donors eyes is utilized?

- (a) Iris (b) Lens
(c) Cornea (d) Retina

M.P.C.S. (Pre) 2004

I.A.S. (Pre) 1999, 2001

Ans. (c)

The cornea is the clear, transparent layer in front of the 'black portion' of the eye. This part of the eye is used in an eye donation and transplant. It is also the main focusing surface, which converges light rays as they enter the eye to focus on the retina. It is thus the most important part of the optical apparatus of the eye. Loss of transparency directly results in loss of vision.

141. A man feels unpleasant due to humidity and heat. The reason for this is :

- (a) More sweating
- (b) Less sweating
- (c) Sweat does not evaporate due to humidity
- (d) Sweat is not formed due to humidity

U.P.P.C.S. (Pre) 1995

Ans. (c)

Man feels unpleasant due to humidity and heat because the body - sweat does not evaporate due to humidity. After the evaporation of sweat, we feel cold.

142. Which of the following actions is related with the evaporation of sweat ?

- (a) Exothermic Action
- (b) Endothermic Action
- (c) Chemical Action
- (d) Salty Reaction

Uttarakhand P.C.S. (Pre) 2010

Ans. (b)

Evaporation of sweat is a physical endothermic process or action. In this process, body's heat is evaporated in the form of sweat. It means in this process heat is absorbed from the body cooling it down. The sweat (produced on the surface of the skin) absorbs heat from the skin to evaporate, thereby creating a cooling effect. Sweating is not an exothermic reaction because it involves physical change only.

143. During dehydration, the substance that is usually lost by the body is :

- (a) Sugar
- (b) Calcium phosphate
- (c) Sodium chloride
- (d) Potassium chloride

U.P.P.C.S. (Mains) 2009

U.P.P.C.S. (Pre) 1998

Ans. (c)

Dehydration is a condition of excessive loss of body fluids. Dehydration is the loss of water and salts (mainly sodium chloride) essential for normal body function.

144. A dehydrated person should not drink :

- (a) Tea
- (b) Coffee
- (c) Sea water
- (d) Alcohol

U.P. Lower Sub. (Pre) 2015

Ans. (c)

Except for sea water, any of the liquid can be used to overcome the lack of water in dehydrated person. The caffeine presented in tea or coffee is diuretic and its consumption causes loss to water in the human body but it is quite low in comparison of water absorbed by tea or coffee. Alcohol is also diuretic but its effect is very low in case of fluid balance in human body. Even alcoholic drinks like light beer which contain less Percentage of alcohol can be used for prevention of dehydration. But it is not with sea water. The human body's cells are at risk of becoming dehydrated on consumption of sea water. UPPSC had mentioned option (d) alcohol as correct answer in its answer key which is wrong. The correct answer should be option (c) i.e. sea water.

145. What percentage of alcohol in human blood causes death?

- (a) 2.0
- (b) 3.0
- (c) 5.0
- (d) 7.0

Jharkhand P.C.S. (Pre) 2016

Ans. (*)

The concentration of alcohol in human blood is called BAC (Blood Alcohol Concentration), which is measured in percentage. 0.10 percent of the quantity means that 0.10 g of alcohol is present in per 100 ml of blood in human body. More than 0.45 percent of BAC in human body can be the reason to death. Option (c) has been marked as the correct answer in JKPSC's answer key.

146. In which faculty is scientist Evan Pavlov known for?

- (a) Physics
- (b) Mathematics
- (c) Chemistry
- (d) Experimental Psychology

38th B.P.S.C. (Pre) 1992

Ans. (d)

Ivan Pavlov was a Russian Psychologist whose research on the psychology of digestion led to the development of the first experimental model of learning classical conditioning. Most of his research was gathered studying salivating dogs. He was awarded the Nobel Prize in Physiology or Medicine in 1904.

147. The pitch of the voice of the women is in general –

- (a) higher than that of men
- (b) marginally lower than that of men
- (c) much lower than that of men
- (d) the same as that of men

I.A.S. (Pre) 1994

Ans. (a)

Normally the pitch of women and children are more than the pitch of men, therefore, the voice of women and children are shriller and pleasant. Pitch is that feature of voice by which the voice is said to be grave or shrill and it depends on frequency.

148. Which is not an indicator of climate change?

- (a) Floral indicator
- (b) Cryogenic indicator
- (c) Tectonic indicator
- (d) Long-term change

M.P.P.C.S. (Pre) 2015

Ans. (d)

Long-term change is not an indicator of climate change. Fossil of plants is an example of a floral indicator of climate change. Glaciation is an example of the cryogenic indicator. Continental drift and pole wandering are some examples of the tectonic indicator.

149. Match List-I (Physiological processes) with List-II (Cell organelles) and select the correct answer by using the codes given below :

List-I	List-II
A. Photosynthesis	1. Plasma membrane
B. Mineral uptake	2. Chloroplast
C. Respiration	3. Mitochondria
D. Protein synthesis	4. Ribosomes

Code :

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 1 | 2 | 3 | 4 |
| (b) | 1 | 2 | 4 | 3 |
| (c) | 2 | 1 | 3 | 4 |
| (d) | 2 | 1 | 4 | 3 |

I.A.S. (Pre) 1996

Ans. (c)

The correctly matched lists are as follows :

Photosynthesis	–	Chloroplast
Mineral uptake	–	Plasma membrane
Respiration	–	Mitochondria
Protein synthesis	–	Ribosomes

150. Who discovered that plants and trees also have life?

- (a) Robert Koach
- (b) J.C. Bose
- (c) Benjamin Franklin
- (d) Louis Pasture

M.P.P.C.S. (Pre) 2015

Ans. (b)

Indian scientist Jagdish Chandra Bose discovered that plants and trees also have a life.

151. In cold storages decomposition of fruits and vegetables:

- (a) Ends forever
- (b) Gets stopped for some time
- (c) Remains unaffected
- (d) Gets retarded

Uttarakhand P.C.S. (Pre) 2005

Ans. (d)

Foods and many other commodities can be preserved by storage at low temperature, which retards the activities of microorganisms. Microorganisms are the spoilage agent and consists of bacteria, yeasts and moulds. Low temperature does not destroy those spoilage agents as does high temperature but greatly reduces their activities.

152. Which animal can not digest fibres properly ?

- (a) Goat
- (b) Cow
- (c) Elephant
- (d) Pig

U.P.P.C.S. (Pre) 1991

Ans. (d)

Pig is an omnivorous animal who can't digest fibre properly, while goat, cow and elephant are ruminant animals who have the bacteria's for digesting fibre.

153. What does the philatelist do?

- (a) Collects the coins
- (b) Collects stamps
- (c) Collects the pictures of birds
- (d) Collects the feathers

Jharkhand P.C.S. (Pre) 2010

Ans. (b)

Philatelist collects postage stamps.

154. A biojector is :

- (a) An H.I.V. immunisation syringe
- (b) A bio-plastic injector
- (c) A painless needle injector

(d) A painless needleless injector

U.P.P.C.S. (Pre) 1997

Ans. (d)

The biojector is an innovative, versatile needle-free injection system that has been used to deliver millions of injections in a wide range of healthcare settings.

155. Robert Webster is known for his work associated with which one of the following ?

- (a) Cardiology (b) Influenza virus
(c) HIV/AIDS (d) Alzheimer

I.A.S. (Pre) 2007

Ans. (b)

Robert Gordon Webster is an avian influenza expert. Webster is world's renowned virologist, an expert in the structure and function of influenza virus proteins and the researcher of new vaccines and anti-virals.

156. Who was the first scientist/doctor, who attempted vaccination?

- (a) Joseph Lister
(b) Robert Koch
(c) Elie Metchnikoff
(d) Edward Jenner
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (d)

Edward Jenner (1749 – 1823) was a British physician and scientist who pioneered the concept of vaccines, and created the smallpox vaccine, the world's first vaccine. Edward Jenner is often called 'the father of immunology'. He first attempted vaccination in 1796.

157. Who is the father of 'Immunology'?

- (a) Robert Koch (b) Louis Pasteur
(c) Edward Jenner (d) Landsteiner

M.P.P.C.S. (Pre) 2013

Ans. (c)

See the explanation of above question.

158. Jonas Salk is known for the discovery of –

- (a) Smallpox vaccine
(b) Cholera vaccine
(c) Polio vaccine
(d) None of the above

U.P. U.D.A./L.D.A. (Spl.) (Pre) 2010

Ans. (c)

Jonas Edward Salk was an American medical researcher and virologist. He discovered and developed the first successful polio vaccine. He was born in New York City, attended New York University School of Medicine, later choose to do medical research instead of becoming a practising physician.

159. Match List-I (Scientists) with List-II (Achievements) and select the correct answer using the codes given below the lists :

List-I (Scientists)

A. Arber and Smith

B. Feldmann

C. Mullis

D. Temin and Baltimore

List-II (Achievements)

1. Developed transgenic plants with Agrobacterium T-DNA
2. Discovered endonucleases
3. Discovered reverse transcriptase
4. Discovered polymerase chain reaction

Code :

	A	B	C	D
(a)	2	1	4	3
(b)	1	2	4	3
(c)	2	1	3	4
(d)	1	2	3	4

I.A.S. (Pre) 2002

Ans. (a)

Arber and Smith	–	Discovered restriction endonucleases
Feldmann	–	Developed transgenic plants with Agrobacterium T-DNA
Mullis	–	Discovered polymerase chain reaction
Temin and Baltimore	–	Discovered reverse transcriptase

160. Which of the following pairs is/are correctly matched?

1. Francis Collins : Mapping human genome
2. Sergey Brin : Google Search Engine
3. Jimmy Wales : Wikipedia

Code :

- (a) 1 and 2
(b) 2 and 3
(c) 3 only
(d) 1, 2 and 3

I.A.S. (Pre) 2008

Ans. (d)

Francis Sellers Collins is an American physician-scientist noted for his discoveries of genes associated with a number of diseases and his leadership of the Human Genome Project. He served as director of the National Institutes of Health (NIH) in Bethesda, Maryland. Sergey Mikhaylovich Brin is an American computer scientist and internet entrepreneur who, together with Larry Page, co-founded Google, one of the most profitable Internet companies. Jimmy Donal "Jimbo" Wales is an American internet entrepreneur best known as the co-founder and promoter of the online non-profit encyclopedia Wikipedia and the for-profit Wikia web-hosting company.

161. Which scientist has propounded the 'erosional cycle'?

- (a) Penck (b) Davis
(c) Hutton (d) Dutton

44th B.P.S.C. (Pre) 2000

Ans. (b)

The cycle of erosion was a model for stream erosion and landscape development proposed by William Morris Davis. He gave the concept of the geographical cycle of erosion which recognises the possibility of obliteration of relief or plantation during the life history of a landscape, by the process of erosion.

162. Auxanometer is used –

- (a) to measure the rate of photosynthesis
(b) to measure the rate of growth
(c) to measure rate of osmosis
(d) to measure the loss of energy

U.P.P.C.S. (Mains) 2010

Ans. (b)

An auxanometer is an apparatus used for measuring rate of growth in plants.

163. A person of mixed European and Indian blood in Latin American is called a –

- (a) Mulatto (b) Mestizo
(c) Meiji (d) Mau Mau

I.A.S. (Pre) 1999

Ans. (b)

In Latin America a person of mixed race, especially one having European and Indigenous American (Indian) parentage is called mestizo. The term 'mestizo' (originated from Latin 'mixticius' meaning 'mixed') is the Spanish word for the general process of mixing ancestries.

164. The age of most ancient mountain group is assessed by –

- (a) Potassium-organ process
(b) C₁₄ process
(c) Ra-Siprocess
(d) Uranium-Lead process

39th B.P.S.C. (Pre) 1994

Ans. (d)

Uranium-lead (U-Pb) process is one of the oldest and most refined methods used to determine the age of ancient mountain rocks that formed from about 1 million years ago or above. Radiocarbon dating (also referred to as carbon dating or carbon-14 dating) is a method of determining the age of an object containing organic material (fossils of dead animals and plants) by using the properties of radiocarbon C-14, a radioactive isotope of carbon.

165. Carbon dating is used to determine the age of –

- (a) Fossils (b) Plants
(c) Rocks (d) None of the above

Uttarakhand P.C.S. (Pre) 2006

Ans. (a)

See the explanation of above question.

166. The age of the tree, in years, can be ascertained by:

- (a) Its weight
(b) Its height
(c) The number of its annular rings
(d) Length of its root

U.P.P.C.S. (Pre) 2000

Ans. (c)

The age of the tree, in years, can be ascertained by the number of its annual (or growth) rings in its stem. The number of rings indicates the number of years the tree lived.

167. Consider the statements :

Assertion (A) : Among various types of soils, clay retains highest amount of water.

Reason (R) : Clay has pore spaces of large size.

Select the correct answer using the codes given below :

Codes :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.

(d) (A) is false, but (R) is true.

U.P. R.O./A.R.O. (Pre) 2017

Ans. (c)

Among various types of soils, clay retains highest amount of water because of the presence of small and fine particles. It can have a very high porosity. Because its grains are very small, there are a lot of (small) pore spaces in between them. Hence, assertion (A) is true but reason (R) is false.

168. Alkaline soils can be improved :

- (a) by the use of lime
- (b) by the use of gypsum
- (c) by the use of saline water
- (d) None of the above

Jharkhand P.C.S. (Pre) 2023

Ans. (b)

Alkaline soils are reclaimed or improved by leaching, crop rotation, providing subsurface drainage area, the addition of gypsum etc. Gypsum application is one of the common amendments used in the reclamation of alkaline soil in India. Quantity of gypsum to be applied depends upon the extent of soil alkalinity, soil texture and crop to be grown.

169. Due to some reasons, if there is a huge fall in the population of species of butterflies, what could be its likely consequence/consequences?

1. **Pollination of some plants could be adversely affected.**
2. **There could be a drastic increase in the fungal infections of some cultivated plants.**
3. **It could lead to a fall in the population of some species of wasps, spiders and birds.**

Select the correct answer using the code given below :

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

I.A.S. (Pre) 2017

Ans. (c)

Butterflies play a vital role in the pollination of plants and the production of crops by transporting pollen grains from one place to another. Thus, huge fall in the population of species of butterflies could adversely affect the pollination of some plants. Hence, statement 1 is correct. Butterflies also act as a lower member of the food chain. There are many different creatures that make butterflies part of their diet. Some of the common predators of butterflies include : wasps, ants, parasitic flies, birds, spiders, snakes, toads, rats, lizards, dragonflies, etc. Thus, huge fall in the

population of butterflies could lead to a fall in the population of some species of wasps, spiders and birds. Hence, statement 3 is also correct.

Statement 2 is incorrect because the drastic increase in the fungal infections of cultivated plants cannot correlated with the fall in the population of butterflies.

170. World's most problematic aquatic weed is :

- (a) Eichhornia
- (b) Trapa
- (c) Wolffia
- (d) Azolla

Jharkhand P.C.S. (Pre) 2021

Ans. (a)

Eichhornia, commonly called water hyacinths, is the world's most problematic aquatic weed. It is also called 'Terror of Bengal'. This plant spreads quickly to create enormous monotypic stands in lakes, rivers, and rice paddies when it comes into contact with the favourable environmental circumstances. Because of its very high rate of growth, it can cover the surface of whole water bodies, choke them and cause a threat to other aquatic life forms.

171. Estuaries possess distinct blooms of excessive growth of a pigmented dinoflagellates. These blooms are called:

- (a) Red tides
- (b) Sea tides
- (c) Black tides
- (d) Sea flowers

I.A.S. (Pre) 1998

Ans. (a)

Red tide is a marine environmental event where protists, including algae and dinoflagellates, go through a tremendous growth period, called a bloom, or an algal bloom. This growth may cause the water to change colours (often red), causing what is commonly known as red tide. Coastal and estuary waters are sometimes affected by algal blooms. These algal blooms may produce toxins or consume dissolved oxygen, and, therefore, cause damage or death to marine life.

172. Consider the following statements :

1. **The common blue-green algae, Spirogyra and Ulothrix, are found in both freshwater ponds and oceans.**
2. **The chameleon can look ahead with one eye, and at the same time look behind with another.**

Which of these statements is are correct ?

- (a) Only 1
- (b) Only 2
- (c) Both 1 and 2
- (d) Neither 1 nor 2

I.A.S. (Pre) 2003

Ans. (b)

Spirogyra is a green algae which contains chlorophyll. In the presence of photosynthesis, they make their own food. They are also known as pond silk or blanket weed. They are mostly found in floating surface of freshwater streams and ponds. It is not found in oceans. Ulothrix is a genus of non-branching filamentous green algae, generally found in fresh and marine water. The chameleon is a species of lizard. It has unique eyes, which can move separately from each other and achieve a visual field of 180 degrees. In this way, it can watch in two different directions at the same time and detect the object on the opposite sides. It has an ability to change the colour of skin.

173. Mycorrhizal biotechnology has been used in rehabilitating degraded sites because mycorrhiza enables the plants to

1. Resist drought and increase absorptive area
2. Tolerate extremes of pH
3. Resist disease infestation

Select the correct answer using the codes given below

- (a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2013

Ans. (d)

A mycorrhiza is a symbiotic (generally mutualistic, but occasionally weakly pathogenic) association between a fungus and the roots of a vascular plant. In a mycorrhizal association, the fungus colonizes the host plant's roots, either intracellularly as in arbuscular mycorrhizal fungi (AMF or AM), or extracellularly as in ectomycorrhizal fungi. They are an important component of soil life and soil chemistry. Mycorrhiza resists drought and increase absorptive area, tolerate extremes of pH and resist disease infestation.

174. 'Biorock technology' is talked about in which one of the following situations?

- (a) Restoration of damaged coral reefs
- (b) Development of building materials using plant residues
- (c) Identification of areas for exploration/extraction of shale gas
- (d) Providing salt licks for wild animals in forests/protected areas

I.A.S. (Pre) 2022

Ans. (a)

Biorock technology is an innovative process originally invented in 1976 by the late German-born architect and marine scientist Wolf Hilbertz to produce natural building materials in the sea. It is a unique method that allows coral

reefs, and other marine ecosystems including seagrass, salt marsh, mangrove, and other oyster reefs to survive and recover from damage caused by excessive nutrients, climate change, and physical destruction etc. Biorock technology is said to be the only sustainable method of protecting coral reefs from mass extinction from global warming.

175. Phytotron is a facility to –

- (a) grow plants under disease-free conditions
- (b) conserve endangered species of plants
- (c) grow plants under controlled conditions
- (d) induce mutations

I.A.S. (Pre) 2000

Ans. (c)

A phytotron is an enclosed research greenhouse used for studying interactions between plants and the environment. Thus, it is a facility to grow plants under controlled conditions.

176. Polyhouse is used for –

- (a) Growing plants
- (b) Storing grain
- (c) Storing vegetables
- (d) None of the above

Uttarakhand P.C.S. (Mains) 2006

Ans. (a)

A polytunnel (also known as a polyhouse, hoop greenhouse or hoop house or high tunnel) is a tunnel made of polyethylene, usually semi-circular, square or elongated in shape. These polyhouses are used in agriculture for growing plants. It protects crops from the bright sun, hails, storms etc.

177. Match List-I with List-II

List-I (Cloth)	List-II (Origin)
A. Linen	1. Coconut plant
B. Coir	2. Flax plant
C. Mohair	3. Feathers of duck and geese
D. Down	4. Angora goat

Select the correct answer using the codes given below.

Code :

	A	B	C	D
(a)	1	3	2	4
(b)	4	3	1	2
(c)	2	1	4	3
(d)	4	1	3	2

69th B.P.S.C. (Pre) 2023

Ans. (c)

The correctly matched lists are as follows :

Cloth		Origin
Linen	–	Flax plant
Coir	–	Coconut plant
Mohair	–	Angora goat
Down	–	Feathers of duck and geese

178. Angora wool is extracted from :

- (a) Rabbit (b) Fox
(c) Goat (d) Sheep
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (a)

‘Angora wool’ (Angora hair or Angora fibre) is extracted from the Angora rabbit. It refers to the downy coat produced by the Angora rabbit. It is distinct from ‘mohair’, which comes from Angora goat.

179. Angora wool is extracted from which of the following animals?

- (a) Sheep (b) Goat
(c) Rabbit (d) Fox

Jharkhand P.C.S. (Pre) 2023

Ans. (c)

See the explanation of above question.

180. Angora wool is obtained from a breed of –

- (a) Sheep (b) Fox
(c) Rabbit (d) Tibetan Deer

Uttarakhand P.C.S. (Mains) 2002

U.P.P.C.S. (Mains) 2010

Ans. (c)

See the explanation of above question.

181. Some marine animals and insects emit light from their body in the darkness. This phenomenon is known as

- (a) Phosphorescence (b) Bioluminescence
(c) Both (a) and (b) (d) None of these

U.P.P.C.S. (Pre) 2014

Ans. (b)

Bioluminescence is light produced by a chemical reaction with in a living organism. The chemical reaction that results in bioluminescence requires two unique chemicals : luciferin and either luciferase or photoprotein. Luciferin is the compound that actually produces light. Most bioluminescent organisms are found in the ocean. These bioluminescent marine species include fish, bacteria and jellies. Phosphorescence is a process in which energy that is absorbed by a substance is released relatively slowly in the form of light.

182. High yielding advanced seeds were used in ‘Green Revolution’, for which the necessary is –

- (a) Less fertilizer and less water
(b) More fertilizer and less water
(c) Less fertilizer and more water
(d) More fertilizer and more water

U.P. U.D.A./L.D.A. (Pre) 2001

Ans. (d)

‘Green Revolution’ is related to the increasement in the food production of crops in large scale in developing countries. In green revolution, to increase the productivity high yielding advanced seeds were used, for which more fertilizers and more water was necessary.

183. Match List-I with List-II and select the correct answer by using the codes given below :

List-I	List-II
A. Green Revolution	1. Oil Seeds
B. White Revolution	2. Food grains
C. Yellow Revolution	3. Fish and Aquaculture
D. Blue Revolution	4. Milk and milk products

Code :

A	B	C	D
(a) 2	1	4	3
(b) 2	4	1	3
(c) 3	4	1	2
(d) 1	2	3	4

U.P.P.C.S. (Mains) 2011

Ans. (b)

The correctly matched lists are as follows :

Green Revolution	–	Food grains
White Revolution	–	Milk and milk Products
Yellow Revolution	–	Oil seeds
Blue Revolution	–	Fish and Aquaculture

184. Which was the principal plant used in Green Revolution?

- (a) Japonica Rice
(b) Indian Rice
(c) Emmer Wheat
(d) Mexican Wheat

Uttarakhand P.C.S. (Pre) 2010

Ans. (d)

The principal crop used in Green Revolution was Mexican wheat.

185. 'Dapog' method of rice nursery was developed in -

- (a) China (b) Indonesia
(c) Japan (d) Philippines

U.P.P.C.S. (Mains) 2010

Ans. (d)

'Dapog' method of raising rice nursery was developed in Philippines. It has been introduced in India from Philippines. The main merit of this method is that less area is needed to raise seedlings. About 25 to 30 square metres of the area is enough to raise seedlings sufficient for planting one hectare.

186. First dwarf rice variety developed in India was -

- (a) IR-8 (b) Jaya
(c) Padma (d) Ratna

U.P.P.C.S. (Mains) 2010

Ans. (b)

The first dwarf variety of rice developed in India was Jaya.

187. Insectivorous fish used for mosquito control is :

- (a) Hilsa (b) Labeo
(c) Gambusia (d) Mystus

U.P.P.C.S (Pre) 2011

Ans. (c)

Gambusia is the large genus of fish in family Poeciliidae (order Cyprinodontiformes). The release of Gambusia into ponds and wells helps in controlling the mosquitoes. Gambusia is known for controlling mosquito populations or mosquito-borne diseases. Gambusia fish is about 3-5 cm in length size, who eats the mosquitoes larva and eggs, which helps in controlling the mosquitoes diseases like dengue and malaria.

188. The release of which one of the following into ponds and wells helps in controlling the mosquitoes?

- (a) Crab (b) Dogfish
(c) Gambusia fish (d) Snail

I.A.S. (Pre) 2008

Ans. (c)

See the explanation of above question.

189. Active acquired immunity is the result of the production of -

- (a) Antibodies (b) Vaccine
(c) Serum (d) Virus that can be executed

R.A.S./R.T.S. (Pre) 1994

Ans. (a)

Active acquired immunity is the result of the production of antibodies. The antibodies, also called immunoglobulin, are protective proteins produced by the immune system which fight against the infection from pathogens such as bacteria and viruses. Antibodies are produced by specialized white blood cells called β -lymphocytes (or B cells).

190. Consider the following as objectives of the New Science and Technology Policy, 2003 and state which is/are correctly stated, with the help of codes given below :

1. Optimal utilisation of existing physical and knowledge resources.
2. Development of innovative technologies.
3. Development of systems and technologies for mitigation and management of natural hazards.
4. Management of intellectual property.

Codes :

- (a) 1 and 2 only (b) 1, 2 and 3 only
(c) 1, 3 and 4 only (d) All the four

U.P.P.C.S. (Pre) 2008

Ans. (d)

Science and Technology Policy, 2003 was launched on January 3, 2003, for expanding new dimensions of innovation and preparation of futuristic perspective of science and technology, during 90th National Science Congress in Bengaluru. The policy includes optimal utilisation of existing physical and knowledge resources, development of innovative technologies, development of system and technologies for mitigation and management of natural hazards and management of intellectual property etc.

191. The first country to impose ban on the sale of tobacco and its products is :

- (a) Sri Lanka (b) Norway
(c) Bhutan (d) Myanmar

Uttarakhand P.C.S. (Pre) 2005

Ans. (c)

The first country to impose a ban on the sale of tobacco and its products is Bhutan in December, 2004.

192. Which of the following planets may be suitable for existence of life?

- (a) Mars (b) Mercury
(c) Venus (d) Jupiter

U.P. P.C.S. (Pre) 2018

Ans. (a)



In the given planets Mars may be suitable for existence of life. The possibility of life on Mars is a subject of significant interest to astrobiology due to its proximity and similarities to Earth. The search for evidence of habitability and organic compounds on Mars is going on.

193. Development of the natural systems is described as :

- (a) Function of the systems
- (b) Evolution of the systems
- (c) Self-sustained process of the systems
- (d) None of the above

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

The natural system regulates natural functions. So, the development of the natural system is defined as self-sustained process of the system.

194. Scientists of the Chandra Shekhar Azad University of Agricultural and Technology have developed a substance that can reduce the percentage of flower droppings in pulses so as to enhance pulse production. Which is that substance ?

- (a) An insecticide called TIVA
- (b) A fertilizer called TIVA
- (c) A nutrient mixture called TIVA
- (d) A growth hormone called TIVA

U.P.P.C.S. (Mains) 2007

Ans. (*)

TIVA is short form of total intravenous anaesthesia. It is a technique of general anaesthesia using a combination of agents given solely by the intravenous route and in the absence of all inhalational agents including nitrous oxide.

195. The true justified belief is often called :

- (a) Hypothesis
- (b) Intelligence
- (c) Knowledge
- (d) None of the above

M.P. P.C.S. (Pre) 2021

Ans. (c)

The true justified belief is often called knowledge. There are three components to the traditional ('tripartite') analysis of knowledge. According to this analysis, justified true belief is necessary and sufficient for knowledge. The tripartite analysis of knowledge is often abbreviated as the 'JTB' analysis, for 'justified true belief'. Much of the twentieth-century literature on the analysis of knowledge took the JTB analysis as its starting point.

196. Consider the following statements :

1. Scientific investigation depends on commitment to practical ideals.

2. Honesty is particularly important in science.

Select the correct answer from the codes given below:

- (a) Both statements 1 and 2 are correct.
- (b) Both statements 1 and 2 are incorrect.
- (c) Statement 1 is correct, 2 is not correct.
- (d) Statement 1 is correct, 2 is unpredictable.

U.P.P.C.S. (Pre) 2024

Ans. (a)

Scientific investigation relies on a commitment to practical ideas and the systematic application of the scientific method, which involves observation, experimentation, and analysis to test and refine hypotheses. Scientific investigation is not just about theoretical concepts; it's about applying those concepts to real-world observations and problems. Honesty is crucial in science for building trust, ensuring accurate knowledge, and fostering scientific progress by promoting transparency and integrity in research and data reporting. Honesty ensures that the scientific process remains focused on seeking truth, rather than manipulating results or suppressing evidence that contradicts existing beliefs. Hence, both statements are correct.

197. Knowledge that we gain from school is called :

- (a) Procedural memory
- (b) Semantic memory
- (c) Declarative memory
- (d) Episodic memory

U.P. P.C.S. (Mains) 2017

Ans. (b)

Semantic memory contains all of the general knowledge about our world that we accumulated throughout our lives. General knowledge such as facts, ideas, meanings and concepts related to our world are stored in semantic memory. It is a subset of long-term memory. So, the knowledge we gain from school is included in semantic memory.

198. Mental activity, that goes on in the brain when a person is processing information, is called :

- (a) Mentation
- (b) A concept
- (c) Thinking
- (d) Mental imagery

U.P. P.C.S. (Mains) 2017

Ans. (c)

Mental activity that goes on in the brain when a person is organizing and attempting to understand information and communicating information to others is called thinking. Thinking exists as the top mental activity demonstrated by man. All human accomplishments and advancement come as the result of thought. Civilization, knowledge, science and technology arise from the thinking process, thought and activity is inseparable.